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UPPER SIXTH — PURE MATHEMATICS WITH STATISTICS

CHAPTER 3

VECTORS

SUGGESTED TIME: 11 PERIODS

IN THIS CHAPTER

This chapter extends the vector work of Lower Sixth to planes in three-dimensional space: their vector, parametric and Cartesian equations, distances, angles and intersections. It provides the complete geometric toolkit needed for the 3-D geometry questions of the GCE Advanced Level examination.

- By the end of this chapter I can write the vector equation r·n = d and the Cartesian equation ax + by + cz = d of a plane from a point and a normal vector.
- By the end of this chapter I can find the Cartesian equation of a plane through three non-collinear points.
- By the end of this chapter I can convert between the parametric (vector) form and the Cartesian form of the equation of a plane.
- By the end of this chapter I can calculate the perpendicular distance from a point to a plane and the distance between two parallel planes.
- By the end of this chapter I can find the acute angle between two planes and the angle between a line and a plane.
- By the end of this chapter I can determine the point of intersection of a line and a plane, including the cases where the line is parallel to or lies in the plane.

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THIS SIGNED COURSE CONTAINS

COURSE complete, illustrated, step by step

METHODS and worked examples

EXERCISES exam style + solutions

SUMMARY SHEET the essentials to remember

CONTENTS

1. Equations of a Plane: Vector and Cartesian Forms
2. Plane Through Three Points and Parametric Form
3. Perpendicular Distance from a Point to a Plane
4. Angles Involving Planes and Lines
5. Intersection of a Line and a Plane; Intersection of Two Planes
6. Families of Planes Through an Intersection and Synthesis
7. Integration activity
8. Methods and worked examples
9. Exercises
10. Solutions to the exercises
11. Summary sheet

CHAPTER OBJECTIVES

- By the end of this chapter I can write the vector equation $\mathbf{r}\cdot\mathbf{n} = d$ and the Cartesian equation $ax + by + cz = d$ of a plane from a point and a normal vector.
- By the end of this chapter I can find the Cartesian equation of a plane through three non-collinear points.
- By the end of this chapter I can convert between the parametric (vector) form and the Cartesian form of the equation of a plane.
- By the end of this chapter I can calculate the perpendicular distance from a point to a plane and the distance between two parallel planes.
- By the end of this chapter I can find the acute angle between two planes and the angle between a line and a plane.
- By the end of this chapter I can determine the point of intersection of a line and a plane, including the cases where the line is parallel to or lies in the plane.
- By the end of this chapter I can find the line of intersection of two non-parallel planes in vector form.
- By the end of this chapter I can write the equation of the family of planes through the line of intersection of two planes and select the member satisfying an extra condition.
- By the end of this chapter I can solve GCE-style problems combining several of these skills.

PREREQUISITES

- Vector algebra from Lower Sixth: position vectors, addition, scalar multiplication, magnitude of a vector
- The scalar (dot) product and its use to find angles and test perpendicularity
- The vector and Cartesian equations of a line in 3-D, including parametric form
- Solving simultaneous linear equations in two and three unknowns
- Basic coordinate geometry in 3-D: coordinates of points, distance between two points

1 EQUATIONS OF A PLANE: VECTOR AND CARTESIAN FORMS

When engineers designed the Douala International Airport runway, they had to ensure that the landing surface was perfectly planar and that the approach path of each aircraft met the runway at a safe, precisely calculated angle. A mason in Bamenda checking whether a wall is perpendicular to the floor, or a carpenter in Yaoundé cutting a roof panel to fit between two sloping faces, is unknowingly solving problems about planes, lines and angles in three-dimensional space. How can we describe a flat surface with an equation, measure how far a point is from it, and find where a line pierces it? This chapter gives you the vector and Cartesian tools to answer exactly these questions — the same tools that appear every year in the GCE Advanced Level Paper 2.

In this first section we answer the most fundamental question: how do we write down the equation of a plane? You already know from Lower Sixth that a line in space is fixed by one point and one direction. We shall see that a plane is fixed by one point and one perpendicular direction, called the **normal vector**.

1.1 What fixes a plane? Intuition and the normal vector

Take a flat sheet of paper and hold it in the air. It can slide, tilt and rotate: it is not yet "fixed". Now pin one point A of the sheet to a fixed position. The sheet can still spin around A in infinitely many ways. Finally, decide the orientation of the sheet by specifying the direction perpendicular to it — imagine a nail driven through A at right angles to the paper. Once the nail's direction is chosen, the sheet cannot move at all. This simple experiment reveals the key fact:

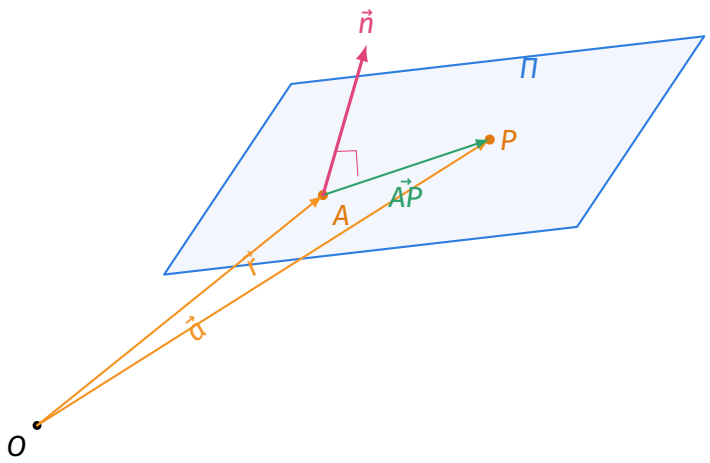
A plane is completely determined by one of its points and the direction perpendicular to it.

DEFINITION **Normal vector of a plane**

A **normal vector** to a plane Π is any non-zero vector $\vec{n}$ which is perpendicular to every line lying in Π. Equivalently, $\vec{n}$ is perpendicular to every vector joining two points of Π.

WARNING **The normal is not unique**

If $\vec{n}$ is normal to a plane, then so is $k\vec{n}$ for any non-zero scalar k: the vectors $\begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$ and $\begin{pmatrix} -2 \\ -4 \\ 2 \end{pmatrix}$ are normal to the same plane. What matters is the direction of the normal, not its length. In particular a plane has infinitely many normal vectors, but they are all parallel to each other.



A plane Π through the point A with position vector $\vec{a}$. The vector $\vec{n}$ is normal to Π; for any point $P(\vec{r})$ of the plane, $\vec{AP}$ lies in Π and is perpendicular to $\vec{n}$.

1.2 Derivation of the vector equation $\vec{r}\cdot\vec{n} = \vec{a}\cdot\vec{n}$

Let Π be the plane through the fixed point A, with position vector $\vec{a} = \vec{OA}$, and with normal vector $\vec{n}$. Take any point P of space, with position vector $\vec{r} = \vec{OP}$. We ask: under what condition on $\vec{r}$ does P lie on Π?

By the triangle law of vector addition,

$$\vec{AP} = \vec{AO} + \vec{OP} = \vec{r} - \vec{a}.$$

Now P lies on Π if and only if the segment AP lies in the plane, and by the definition of the normal vector this happens if and only if $\vec{AP}$ is perpendicular to $\vec{n}$. Two vectors are perpendicular if and only if their scalar product is zero, so

$$P \in \Pi \iff \vec{n}\cdot\vec{AP} = 0 \iff \vec{n}\cdot(\vec{r}-\vec{a}) = 0 \iff \vec{n}\cdot\vec{r} = \vec{n}\cdot\vec{a}.$$

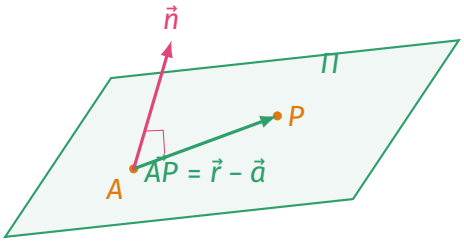
Since $\vec{a}$ and $\vec{n}$ are fixed, the quantity $\vec{a}\cdot\vec{n}$ is a fixed constant, which we call d. We have proved the central result of this section.

PROPERTY **Vector equation of a plane — proof examinable**

The plane through the point A with position vector $\vec{a}$, with normal vector $\vec{n}$, has vector equation

$$\vec{r}\cdot\vec{n} = \vec{a}\cdot\vec{n} = d,$$

where $\vec{r}$ is the position vector of a general point of the plane and d is a constant. The proof (using $\vec{n}\cdot(\vec{r}-\vec{a}) = 0$) is examinable and is frequently demanded with the instruction "hence" or "show that".



$$\vec{n} \cdot \vec{AP} = 0$$

The key geometric fact: for every point P of the plane, $\vec{AP}$ lies in the plane, so $\vec{n} \cdot \vec{AP} = 0$.

1.3 From the vector form to the Cartesian form $ax + by + cz = d$

Write the vectors in components with respect to the usual orthonormal basis $(\vec{i}, \vec{j}, \vec{k})$:

$$\vec{r} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}, \qquad \vec{n} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}, \qquad \vec{a} = \begin{pmatrix} x_0 \\ y_0 \\ z_0 \end{pmatrix}.$$

Computing the scalar products in $\vec{r} \cdot \vec{n} = \vec{a} \cdot \vec{n}$ gives

$$ax + by + cz = ax_0 + by_0 + cz_0.$$

The right-hand side is a constant; calling it d we obtain the **Cartesian equation** of the plane:

$$ax + by + cz = d.$$

KEY POINTS **Reading the normal from the Cartesian equation**

In the Cartesian equation $ax + by + cz = d$, the coefficients of x, y and z are exactly the components of a normal vector:

$$\vec{n} = a\vec{i} + b\vec{j} + c\vec{k} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}.$$

For example, the plane $2x - 3y + 5z = 7$ has normal vector $\begin{pmatrix} 2 \\ -3 \\ 5 \end{pmatrix}$. The constant d on the right-hand side carries the information about which point the plane passes through.

WARNING **Coefficients give the normal, NOT a point**

A classic GCE error: faced with $2x - 3y + 5z = 7$, candidates claim that $(2, -3, 5)$ is a point of the plane. It is not — it is the normal vector. Check: substituting $x = 2$, $y = -3$, $z = 5$ gives $4 + 9 + 25 = 38 \neq 7$, so the point $(2, -3, 5)$ does not even lie on the plane. The coefficients (a, b, c) always describe a direction perpendicular to the plane, never a point on it.

1.4 Finding the equation of a plane: the standard method

METHOD **Equation of a plane from a point and a normal**

To find the equation of a plane:

1. Identify a point $A(x_0, y_0, z_0)$ on the plane.
2. Identify a normal vector $\vec{n} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$ (given directly, or obtained from a line perpendicular to the plane, or from the normal of a parallel plane).
3. Compute $d = \vec{a} \cdot \vec{n} = ax_0 + by_0 + cz_0$.
4. Write the equation $\vec{r} \cdot \vec{n} = d$ (vector form) or $ax + by + cz = d$ (Cartesian form).
5. Check: substitute the coordinates of A into your equation — it must be satisfied.

WORKED EXAMPLE 1 **Plane through a given point with a given normal**

Find the vector and Cartesian equations of the plane Π passing through the point $A(1, -2, 3)$ and perpendicular to the vector $\vec{n} = 2\vec{i} + \vec{j} - 4\vec{k}$.

SOLUTION

Solution. We have a point and a normal, so we apply the method. With $\vec{a} = \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix}$ and $\vec{n} = \begin{pmatrix} 2 \\ 1 \\ -4 \end{pmatrix}$:

$$d = \vec{a} \cdot \vec{n} = (1)(2) + (-2)(1) + (3)(-4) = 2 - 2 - 12 = -12.$$

Vector equation:

$$\vec{r} \cdot \begin{pmatrix} 2 \\ 1 \\ -4 \end{pmatrix} = -12.$$

Cartesian equation: writing $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$,

$$2x + y - 4z = -12.$$

Check: at $A(1, -2, 3)$, $2(1) + (-2) - 4(3) = 2 - 2 - 12 = -12$. ✓ The point lies on the plane, as required.

WORKED EXAMPLE 2 **Plane perpendicular to a given line**

A line L has vector equation $\vec{r} = \begin{pmatrix} 3 \\ 0 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix}$. Find the Cartesian equation of the plane which is perpendicular to L and passes through the point $B(2, 1, 4)$.

SOLUTION

Solution. The plane is perpendicular to L, so the direction vector of L is a normal vector of the plane:

$$\vec{n} = \begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix}.$$

Then

$$d = \vec{n} \cdot \vec{OB} = (1)(2) + (-2)(1) + (2)(4) = 2 - 2 + 8 = 8.$$

Hence the plane has equation

$$x - 2y + 2z = 8.$$

Check: at $B(2, 1, 4)$, $2 - 2 + 8 = 8$. ✓

1.5 Parallel planes and planes perpendicular to a line

Two planes that never meet — like the floor and the ceiling of a classroom — have exactly the same orientation, which means their normals point in the same (or opposite) direction.

PROPERTY **Parallel planes — exam tip**

Two planes are **parallel** if and only if their normal vectors are parallel, i.e. one normal is a scalar multiple of the other. Thus $a_1x+b_1y+c_1z = d_1$ and $a_2x+b_2y+c_2z = d_2$ are parallel exactly when $(a_2, b_2, c_2) = k(a_1, b_1, c_1)$ for some non-zero k. Consequently, any plane parallel to $ax + by + cz = d$ has equation $ax + by + cz = d'$ for some new constant d', found using a point of the required plane. In the examination, state this justification explicitly to earn the method mark.

EXEMPLE **Plane parallel to a given plane**

Find the equation of the plane parallel to $\Pi : 3x - y + 2z = 5$ and passing through $C(0, 2, -1)$.

Solution. A parallel plane has the same normal $\vec{n} = \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}$, so its equation is $3x - y + 2z = d'$.

Substituting C:

$$d' = 3(0) - (2) + 2(-1) = -4.$$

Hence the required plane is $3x - y + 2z = -4$.

1.6 Checking whether a point lies on a plane

A point $M(x_1, y_1, z_1)$ lies on the plane $ax + by + cz = d$ if and only if its coordinates satisfy the equation, that is $ax_1+by_1+cz_1 = d$. In vector language, M lies on the plane $\vec{r}\cdot\vec{n} = d$ if and only if $\vec{OM}\cdot\vec{n} = d$.

EXEMPLE **Testing points on a plane**

Does the point $M(2, 0, 3)$ lie on the plane $\Pi : 2x + y - 4z = -12$? What about $N(1, 1, 1)$?
Solution. For M: $2(2) + 0 - 4(3) = 4 - 12 = -8 \neq -12$, so $M \notin \Pi$.
For N: $2(1) + 1 - 4(1) = -1 \neq -12$, so $N \notin \Pi$ either. However $A(1, -2, 3)$ gives -12, so $A \in \Pi$ (as we built it).

WARNING **Common mistakes to avoid**

- Forgetting that $\vec{n}$ must be non-zero: $0x + 0y + 0z = d$ does not define a plane.
- Mixing up signs when reading the normal: the normal of $x - 2y + 2z = 8$ is $(1, -2, 2)$, not $(1, 2, 2)$.
- Computing d with the wrong point, or with the normal vector instead of the position vector of the given point.

- Giving $ax + by + cz = 0$ systematically: the right-hand side is 0 only when the plane passes through the origin.

DID YOU KNOW? **Planes in the Cameroonian landscape**

The idea of describing a flat surface by a point and a perpendicular direction is used daily by surveyors mapping the slopes of Mount Cameroon and by civil engineers levelling the terraces of the Bamenda highlands. In geodesy, the "local horizontal plane" at a site is defined precisely by the vertical (the normal direction given by a plumb line) at one reference point — exactly our data $(A, \vec{n})$.

EXERCISE 1 **Practice**

1. Find the vector and Cartesian equations of the plane through $A(2, -1, 5)$ with normal vector $\vec{n} = \begin{pmatrix} 3 \\ 2 \\ -1 \end{pmatrix}$.
2. The line $\vec{r} = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$ is perpendicular to a plane Π which contains the origin. Find the Cartesian equation of Π.
3. Find the equation of the plane parallel to $4x + y - 3z = 7$ and passing through $P(1, 1, 1)$.
4. Determine which of the points $Q(3, -1, 2)$ and $R(0, 4, -1)$ lie on the plane $x + 2y - z = -1$.

SOLUTION **Exercise 1**

1. $d = 3(2)+2(-1)-1(5) = 6-2-5 = -1$. Vector form: $\vec{r}\cdot\begin{pmatrix} 3 \\ 2 \\ -1 \end{pmatrix} = -1$; Cartesian form: $3x+2y-z = -1$.
2. Normal $\vec{n} = (2, -1, 3)$; plane through the origin so $d = 0$: $2x - y + 3z = 0$.
3. Same normal: $4x + y - 3z = d'$ with $d' = 4 + 1 - 3 = 2$. Equation: $4x + y - 3z = 2$.
4. For Q: $3 + 2(-1) - 2 = -1$, so Q lies on the plane. For R: $0 + 8 + 1 = 9 \neq -1$, so R does not.

KEY POINTS **Summary of Section 1**

- A plane is fixed by one point $A(\vec{a})$ and one normal vector $\vec{n}$.
- Vector equation: $\vec{r} \cdot \vec{n} = \vec{a} \cdot \vec{n} = d$, derived from $\vec{n} \cdot (\vec{r} - \vec{a}) = 0$.
- Cartesian equation: $ax + by + cz = d$, where (a, b, c) are the components of the normal.
- Parallel planes $\iff$ parallel normals; a parallel plane keeps the same left-hand side, only d changes.
- A point lies on the plane if and only if its coordinates satisfy the equation.

2 PLANE THROUGH THREE POINTS AND PARAMETRIC FORM

In the previous section we learnt that a plane is completely determined by one of its points A and a **normal vector** $\vec{n}$, and we wrote its equation in the vector form $(\vec{r} - \vec{a}) \cdot \vec{n} = 0$ and in the Cartesian form $ax + by + cz = d$. But in many problems — and very often at the GCE — you are not given a normal vector. Instead you are given three points of the plane, or two directions lying in the plane. The purpose of this section is to show how the same machinery (dot product, normal vector, Cartesian equation) handles these situations, and to introduce a second, very flexible way of describing a plane: the **parametric vector equation** $\vec{r} = \vec{a} + \lambda\vec{u} + \mu\vec{v}$.

2.1 The Plane Determined by Three Non-Collinear Points

Intuition. Place three fingertips of your hand on a table so that they are not in a straight line: the table top is the only flat surface that can touch all three fingertips at once. But if the three fingertips are in a straight line, you can tilt a book around that line and still touch all three — infinitely many planes pass through them. This is exactly the mathematical situation.

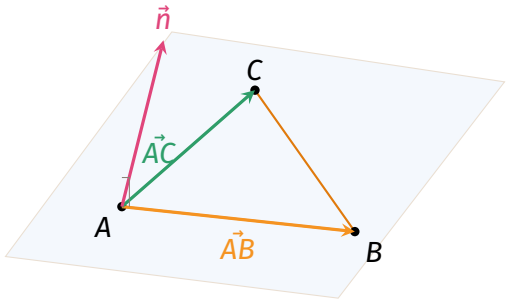
PROPERTY **Existence and uniqueness**

Through three **non-collinear** points A, B, C there passes exactly one plane. If the three points are collinear, infinitely many planes contain them.

Strategy. Since A, B, C lie in the plane, the vectors $\vec{AB}$ and $\vec{AC}$ lie in the plane, and they are not parallel (precisely because A, B, C are not collinear). A normal vector $\vec{n}$ must be perpendicular to every vector of the plane, in particular to both $\vec{AB}$ and $\vec{AC}$. Using the dot product, this gives two equations:

$$\vec{n} \cdot \vec{AB} = 0 \qquad \text{and} \qquad \vec{n} \cdot \vec{AC} = 0.$$

These two equations determine $\vec{n}$ up to a scalar multiple (two equations, three unknowns), which is all we need: any non-zero multiple of a normal vector is still a normal vector.



Three non-collinear points A, B, C determine a unique plane. The vectors $\vec{AB}$ and $\vec{AC}$ lie in the plane; the normal $\vec{n}$ is perpendicular to both.

METHOD **Cartesian equation of the plane through three points**

1. Check non-collinearity: compute $\vec{AB}$ and $\vec{AC}$ and verify that one is not a scalar multiple of the other.
2. Find a normal: let $\vec{n} = \begin{pmatrix} p \\ q \\ r \end{pmatrix}$ and solve simultaneously $\vec{n} \cdot \vec{AB} = 0$ and $\vec{n} \cdot \vec{AC} = 0$. Express two of p, q, r in terms of the third (the ratio method), then choose a convenient value to get integer components.
3. Find d: write the equation as $px + qy + rz = d$ and substitute the coordinates of any one of the three points to compute d.
4. Check: verify that the other two points also satisfy the equation.

WORKED EXAMPLE 3 **Plane through three points**

Find the Cartesian equation of the plane passing through $A(1, 0, 2)$, $B(2, 1, 1)$ and $C(-1, 2, 1)$.

SOLUTION

Step 1: direction vectors in the plane.

$$\vec{AB} = \begin{pmatrix} 2-1 \\ 1-0 \\ 1-2 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}, \qquad \vec{AC} = \begin{pmatrix} -1-1 \\ 2-0 \\ 1-2 \end{pmatrix} = \begin{pmatrix} -2 \\ 2 \\ -1 \end{pmatrix}.$$

These are not proportional (no single scalar k gives $(1, 1, -1) = k(-2, 2, -1)$), so A, B, C are not collinear and define a unique plane.

Step 2: normal vector by the ratio method. Let $\vec{n} = (p, q, r)$. Then

$$\vec{n} \cdot \vec{AB} = 0 : \qquad p + q - r = 0,$$
$$\vec{n} \cdot \vec{AC} = 0 : \qquad -2p + 2q - r = 0.$$

Subtracting the second equation from the first: $3p - q = 0$, so $q = 3p$. Substituting back: $r = p + q = 4p$. Hence

$$p : q : r = 1 : 3 : 4, \qquad \vec{n} = \begin{pmatrix} 1 \\ 3 \\ 4 \end{pmatrix}.$$

Step 3: find d. The equation has the form $x + 3y + 4z = d$. Using $A(1, 0, 2)$: $d = 1 + 0 + 8 = 9$.
Step 4: check. B: $2 + 3 + 4 = 9$ ✓; C: $-1 + 6 + 4 = 9$ ✓.

$$\boxed{x + 3y + 4z = 9}$$

2.2 Lesson 43: The Collinear Case — Three Points on a Line

The syllabus explicitly mentions the case of three collinear points. This is the trap case: if A, B, C are collinear, then $\vec{AB}$ and $\vec{AC}$ are parallel, the equations $\vec{n} \cdot \vec{AB} = 0$ and $\vec{n} \cdot \vec{AC} = 0$ reduce to a single condition, and there are **infinitely many planes** through the three points (any plane containing the line ABC).

WARNING **Three collinear points: the degenerate case**

Always check first that the three given points are not collinear before claiming a unique plane. For example $A(1, 1, 0)$, $B(2, 2, 1)$, $C(3, 3, 2)$ are collinear since $\vec{AB} = (1, 1, 1)$ and $\vec{AC} = (2, 2, 2) = 2\vec{AB}$: no unique plane exists. In an examination, if you detect collinearity, state clearly: "The points are collinear, therefore they do not define a unique plane; infinitely many planes contain the line through these points."

EXEMPLE **Collinear points**

Show that $A(1, 2, 3)$, $B(0, 1, 1)$, $C(2, 3, 5)$ are collinear and explain the consequence. $\vec{AB} = (-1, -1, -2)$, $\vec{AC} = (1, 1, 2) = -\vec{AB}$. The vectors are parallel, so the points are collinear. They lie on a single line; any plane containing that line contains all three points. There is no unique plane.

2.3 The Parametric (Vector) Form of the Equation of a Plane

Intuition. Stand at point A of the plane. To reach any other point P of the plane, walk some dis-

tance along one direction $\vec{u}$ of the plane, then some distance along a second, different direction $\vec{v}$. The two "amounts" λ and μ are your coordinates on the plane, exactly as x and y are coordinates on a map of Douala: two numbers suffice, because a plane is two-dimensional.

DEFINITION **Parametric vector equation of a plane**

Let A be a point of the plane with position vector $\vec{a}$, and let $\vec{u}$ and $\vec{v}$ be two **non-parallel** vectors lying in the plane ($\vec{u} \nparallel \vec{v}$). The plane is the set of points P with position vector

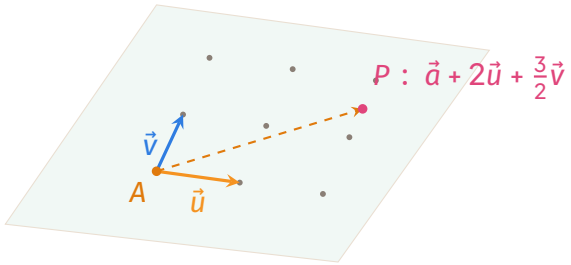
$$\vec{r} = \vec{a} + \lambda\vec{u} + \mu\vec{v}, \qquad \lambda, \mu \in \mathbb{R}.$$

The scalars λ and μ are called **parameters**; $\vec{u}$ and $\vec{v}$ are **direction vectors** of the plane.

For three given points A, B, C, the natural choice is $\vec{u} = \vec{AB}$, $\vec{v} = \vec{AC}$, giving

$$\vec{r} = \vec{OA} + \lambda\,\vec{AB} + \mu\,\vec{AC}.$$

Note that $\lambda = 1, \mu = 0$ gives B; $\lambda = 0, \mu = 1$ gives C; $\lambda = \mu = 0$ gives A.



Parametric form: from A, the two non-parallel directions $\vec{u}$ and $\vec{v}$ generate every point $\vec{a} + \lambda\vec{u} + \mu\vec{v}$ of the plane (grid of dots).

EXEMPLE **Parametric form from three points**

For $A(1, 0, 2)$, $B(2, 1, 1)$, $C(-1, 2, 1)$ as above, a parametric vector equation is

$$\vec{r} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} + \mu \begin{pmatrix} -2 \\ 2 \\ -1 \end{pmatrix}, \qquad \lambda, \mu \in \mathbb{R}.$$

2.4 From Parametric Form to Cartesian Form: Eliminating the Parameters

METHOD **Eliminating λ and μ**

1. Write the three scalar equations $x = \ldots$, $y = \ldots$, $z = \ldots$ in terms of λ and μ.
2. Solve two of the equations simultaneously for λ and μ in terms of x, y, z.
3. Substitute into the third equation and simplify: the result is the Cartesian equation $ax + by + cz = d$.

(Equivalently: find a normal $\vec{n}$ perpendicular to both $\vec{u}$ and $\vec{v}$ by solving $\vec{n} \cdot \vec{u} = 0$, $\vec{n} \cdot \vec{v} = 0$, then proceed as before.)

WORKED EXAMPLE 4 **Parametric to Cartesian**

A plane has equation $\vec{r} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} + \mu \begin{pmatrix} -2 \\ 2 \\ -1 \end{pmatrix}$. Find its Cartesian equation.

SOLUTION

The scalar equations are

$$x = 1 + \lambda - 2\mu, \qquad y = \lambda + 2\mu, \qquad z = 2 - \lambda - \mu.$$

From the first two equations:

$$x + y = 1 + 2\lambda \quad \Rightarrow \quad \lambda = \frac{x + y - 1}{2},$$
$$x - y = 1 - 4\mu \quad \Rightarrow \quad \mu = \frac{1 - x + y}{4}.$$

Substitute into the third equation:

$$z = 2 - \frac{x + y - 1}{2} - \frac{1 - x + y}{4}$$
$$4z = 8 - 2(x + y - 1) - (1 - x + y)$$
$$4z = 8 - 2x - 2y + 2 - 1 + x - y$$
$$4z = 9 - x - 3y.$$

Rearranging gives the Cartesian equation:

$$\boxed{x + 3y + 4z = 9}$$

Verification (normal method): $\vec{n} \cdot \vec{u} = p + q - r = 0$, $\vec{n} \cdot \vec{v} = -2p + 2q - r = 0$ give $q = 3p$, $r = 4p$, so $\vec{n} = (1, 3, 4)$ and equation $x + 3y + 4z = d$. Point $(1, 0, 2)$ gives $d = 9$. The results agree.

WARNING **Check your elimination**

Eliminating parameters by hand is error-prone. Always cross-check: either verify that the three original points satisfy your Cartesian equation, or find the normal from $\vec{n}\cdot\vec{u} = \vec{n}\cdot\vec{v} = 0$ and compare. A single arithmetic slip changes the whole plane.

2.5 From Cartesian Form to Parametric Form

To go the other way, choose two of the variables as parameters and express the third in terms of them. This is always possible because a Cartesian equation $ax+by+cz = d$ with $(a, b, c) \neq (0, 0, 0)$ can be solved for one variable in terms of the other two.

WORKED EXAMPLE 5 **Cartesian to parametric**

Write the plane $x + 3y + 4z = 9$ in parametric vector form.

SOLUTION

Set $y = \lambda$ and $z = \mu$ (free parameters). Then $x = 9 - 3\lambda - 4\mu$, so

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 9-3\lambda-4\mu \\ \lambda \\ \mu \end{pmatrix} = \begin{pmatrix} 9 \\ 0 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -3 \\ 1 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} -4 \\ 0 \\ 1 \end{pmatrix}.$$

Hence $\vec{r} = \begin{pmatrix} 9 \\ 0 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -3 \\ 1 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} -4 \\ 0 \\ 1 \end{pmatrix}$, $\lambda, \mu \in \mathbb{R}$. Check: both direction vectors are perpendicular to $\vec{n} = (1, 3, 4)$: $(-3)(1) + (1)(3) + 0 = 0$ and $(-4)(1) + 0 + (1)(4) = 0$ ✓.

WARNING **The parametric form is not unique**

The same plane has infinitely many parametric forms: any point of the plane can play the role of A, and any two non-parallel directions in the plane can play the roles of $\vec{u}$ and $\vec{v}$. Do not be surprised if your answer differs from a classmate's; check instead that (i) your point satisfies the Cartesian equation and (ii) both your direction vectors are perpendicular to the normal.

DID YOU KNOW? **Exam tip**

Any vector perpendicular to both $\vec{u}$ and $\vec{v}$ can serve as the normal $\vec{n}$. When the ratio method

gives fractional components, scale the vector to convenient small integers: for instance $\vec{n} = \left(\frac{1}{2}, \frac{3}{2}, 2\right)$ should immediately be replaced by $(1, 3, 4)$. A normal vector is a direction, not a fixed length — multiplying it by any non-zero scalar leaves the plane unchanged, and integer components make the computation of d far safer under exam pressure.

KEY POINTS **Key points of this section**

- Three non-collinear points determine a unique plane; three collinear points lie on infinitely many planes — always check first (Lesson 43).
- Method: form $\vec{AB}, \vec{AC}$; solve $\vec{n} \cdot \vec{AB} = \vec{n} \cdot \vec{AC} = 0$ by the ratio method; get d by substituting any of the three points.
- Parametric form: $\vec{r} = \vec{a} + \lambda\vec{u} + \mu\vec{v}, \lambda, \mu \in \mathbb{R}$, with $\vec{u} \nparallel \vec{v}$.
- Parametric → Cartesian: eliminate λ and μ from x, y, z; Cartesian → parametric: choose two variables as parameters.
- The parametric form is never unique; the Cartesian form is unique up to a non-zero multiple.

EXERCISE 2 **Practice**

1. Find the Cartesian equation of the plane through $P(2, -1, 3)$, $Q(1, 1, 1)$ and $R(3, 0, 4)$.
2. Show that the points $A(1, 2, 3)$, $B(0, 1, 1)$ and $C(2, 3, 5)$ are collinear, and explain why they do not define a unique plane.
3. A plane has equation $\vec{r} = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$. Find its Cartesian equation.
4. Write the plane $2x - y + 3z = 6$ in parametric vector form, and verify that your direction vectors are perpendicular to the normal.

SOLUTION **Exercise 1**

$\vec{PQ} = (-1, 2, -2)$, $\vec{PR} = (1, 1, 1)$. With $\vec{n} = (p, q, r)$: $-p + 2q - 2r = 0$ and $p + q + r = 0$. Adding: $3q - r = 0$, so $r = 3q$; then $p = -q - r = -4q$. Take $q = 1$: $\vec{n} = (-4, 1, 3)$. Equation: $-4x + y + 3z = d$; using $P(2, -1, 3)$: $d = -8 - 1 + 9 = 0$. Hence $-4x + y + 3z = 0$, i.e. $4x - y - 3z = 0$. Check Q:

$4 - 1 - 3 = 0$ ✓; R: $12 - 0 - 12 = 0$ ✓.

SOLUTION **Exercise 2**

$\vec{AB} = (-1, -1, -2)$ and $\vec{AC} = (1, 1, 2) = -\vec{AB}$. The vectors are parallel, so A, B, C are collinear. Any plane containing the line ABC contains all three points, and there are infinitely many such planes (rotate a plane about the line): no unique plane is defined.

SOLUTION **Exercise 3**

$x = \lambda + 2\mu$, $y = 1 + \mu$, $z = 2 + \lambda$. From the second: $\mu = y - 1$; from the third: $\lambda = z - 2$. Substituting into the first: $x = (z-2) + 2(y-1) = 2y + z - 4$, so $\boxed{x - 2y - z + 4 = 0}$. Check with normal: $p + r = 0$, $2p + q = 0$; take $p = 1$: $r = -1$, $q = -2$, giving $x - 2y - z = d$; the point $(0, 1, 2)$ gives $d = -4$ ✓.

SOLUTION **Exercise 4**

Set $y = \lambda$, $z = \mu$: then $x = \dfrac{6 + \lambda - 3\mu}{2} = 3 + \dfrac{\lambda}{2} - \dfrac{3\mu}{2}$, so

$$\vec{r} = \begin{pmatrix} 3 \\ 0 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} \frac{1}{2} \\ 1 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} -\frac{3}{2} \\ 0 \\ 1 \end{pmatrix},$$

or, scaling the directions, $\vec{r} = (3, 0, 0) + \lambda(1, 2, 0) + \mu(-3, 0, 2)$. Check: $(1, 2, 0) \cdot (2, -1, 3) = 2 - 2 + 0 = 0$ ✓; $(-3, 0, 2) \cdot (2, -1, 3) = -6 + 0 + 6 = 0$ ✓.

3 PERPENDICULAR DISTANCE FROM A POINT TO A PLANE

In the previous section we learnt how to describe a plane completely: through three non-collinear points, or through a point with two direction vectors, passing from the vector form $\vec{r} = \vec{a} + s\vec{u} + t\vec{v}$ to the Cartesian form $ax + by + cz = d$. We now ask a natural metric question: once a plane is fixed in space, **how far is a given point from it?** This question appears every year in some form at the GCE Advanced Level, and it is also the key to measuring distances between parallel planes, finding feet of perpendiculars, and reflecting points in planes.

3.1 Geometric Meaning of the Perpendicular Distance

EXPERIMENT **The shortest path to a wall**

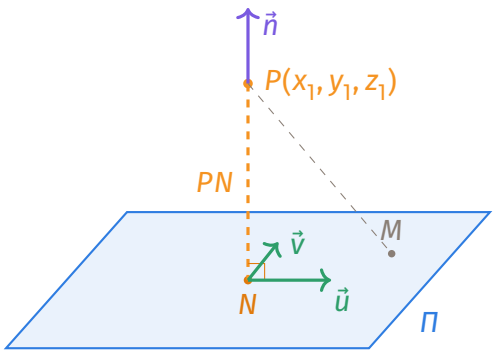
Stand anywhere in your classroom and look at one of the walls. There are infinitely many ways to walk from your position to the wall: you could walk straight towards it, or walk obliquely to the left, or to the right. Common sense tells you that the shortest walk is the one where you go straight to the wall, meeting it at a right angle. Any slanting path is longer. The same is true in three dimensions: among all segments joining a point P to points of a plane Π, the shortest one is the segment that is perpendicular to Π.

DEFINITION **Perpendicular distance from a point to a plane**

Let Π be a plane and P a point not on Π. The line through P perpendicular to Π meets Π at a unique point N, called the **foot of the perpendicular** from P to Π. The **perpendicular distance** from P to Π is the length

$$d(P, \Pi) = PN.$$

It is the shortest distance from P to any point of the plane: for every point M of Π different from N, the triangle PNM is right-angled at N, so by Pythagoras $PM^2 = PN^2 + NM^2 > PN^2$, hence $PM > PN$.



The perpendicular segment PN (parallel to the normal $\vec{n}$) is shorter than any oblique segment PM.

3.2 Derivation of the Distance Formula

Let the plane have equation $\Pi : ax + by + cz = d$, with normal vector $\vec{n} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$, and let $P(x_1, y_1, z_1)$ be any point. Choose any point $A(x_0, y_0, z_0)$ lying **on** the plane, so that

$$ax_0 + by_0 + cz_0 = d.$$

The distance PN is the absolute value of the scalar projection of $\vec{AP}$ onto the unit normal $\hat{n} = \dfrac{\vec{n}}{\|\vec{n}\|}$:

$$PN = \left|\vec{AP} \cdot \hat{n}\right| = \frac{\left|\vec{AP} \cdot \vec{n}\right|}{\|\vec{n}\|}.$$

Now $\vec{AP} = \begin{pmatrix} x_1 - x_0 \\ y_1 - y_0 \\ z_1 - z_0 \end{pmatrix}$, so

$$\vec{AP} \cdot \vec{n} = a(x_1 - x_0) + b(y_1 - y_0) + c(z_1 - z_0) = (ax_1 + by_1 + cz_1) - (ax_0 + by_0 + cz_0) = ax_1 + by_1 + cz_1 - d,$$

since $ax_0 + by_0 + cz_0 = d$. Also $\|\vec{n}\| = \sqrt{a^2 + b^2 + c^2}$. This proves the fundamental formula.

PROPERTY **Perpendicular distance from a point to a plane — proof examinable**

The perpendicular distance from the point $P(x_1, y_1, z_1)$ to the plane $ax + by + cz = d$ is

$$\delta = \frac{\left|ax_1 + by_1 + cz_1 - d\right|}{\sqrt{a^2 + b^2 + c^2}}$$

Proof. Derived above by projecting $\vec{AP}$ onto the normal direction, where A is any point of the plane. The derivation is short and is frequently demanded at the GCE: learn it.

WARNING **Distance is never negative**

The numerator must be an absolute value. If you compute $ax_1+by_1+cz_1-d$ and obtain -7, the distance is $\dfrac{7}{\sqrt{a^2 + b^2 + c^2}}$, not a negative number. Writing a negative distance costs marks at the GCE.

WARNING **Put the equation in standard form first**

The formula applies only when the plane is written as $ax+by+cz = d$ with all constant terms on one side. If the plane is given as $2x - y + 2z + 5 = 0$, rewrite it as $2x - y + 2z = -5$, so $d = -5$, and the numerator becomes $|2x_1 - y_1 + 2z_1 - (-5)| = |2x_1 - y_1 + 2z_1 + 5|$. Mixing up the sign of d is the most common error.

WORKED EXAMPLE 6 **Distance from a point to a plane**

Find the perpendicular distance from the point $P(3, -1, 2)$ to the plane $\Pi : \ 2x + y - 2z = 4$.

SOLUTION

Solution. Here $a = 2, b = 1, c = -2, d = 4$, and $(x_1, y_1, z_1) = (3, -1, 2)$.

$$\text{Numerator: } |2(3) + (-1) - 2(2) - 4| = |6 - 1 - 4 - 4| = |-3| = 3.$$

$$\text{Denominator: } \sqrt{2^2 + 1^2 + (-2)^2} = \sqrt{4 + 1 + 4} = \sqrt{9} = 3.$$

$$\delta = \frac{3}{3} = 1.$$

The point P is at distance 1 unit from the plane.

3.3 Interpreting the Sign of $ax_1 + by_1 + cz_1 - d$

The quantity inside the absolute value carries extra information. The plane $\Pi : ax + by + cz = d$ divides space into two **half-spaces**:

- points with $ax + by + cz - d > 0$ lie on the side towards which $\vec{n}$ points;
- points with $ax + by + cz - d < 0$ lie on the opposite side;
- points with $ax + by + cz - d = 0$ lie on the plane.

So the **sign** tells you which side the point is on, and the **absolute value** tells you how far. Two points are on the same side of the plane exactly when the expressions $ax+by+cz-d$ evaluated at each point have the same sign. This is a quick test examiners like to combine with the distance formula.

EXEMPLE **Same side or opposite sides?**

For the plane $2x + y - 2z = 4$: at $P(3, -1, 2)$ we found $2x_1 + y_1 - 2z_1 - 4 = -3 < 0$; at $Q(4, 1, 1)$ we get $8 + 1 - 2 - 4 = 3 > 0$. Hence P and Q lie on opposite sides of the plane, each at distance 1 from it.

3.4 Distance from the Origin and Distance Between Parallel Planes

Setting $(x_1, y_1, z_1) = (0, 0, 0)$ in the general formula gives an immediate corollary.

PROPERTY **Distance from the origin to a plane**

The distance from the origin O to the plane $ax + by + cz = d$ is

$$\delta = \frac{|d|}{\sqrt{a^2 + b^2 + c^2}} = \frac{|d|}{\|\vec{n}\|}.$$

Two planes are **parallel** exactly when they have the same normal vector, i.e. equations $ax + by +$

$cz = d_1$ and $ax + by + cz = d_2$. To find the distance between them, take any point of one plane and measure its distance to the other. A pleasant shortcut results.

PROPERTY **Distance between two parallel planes — exam tip**

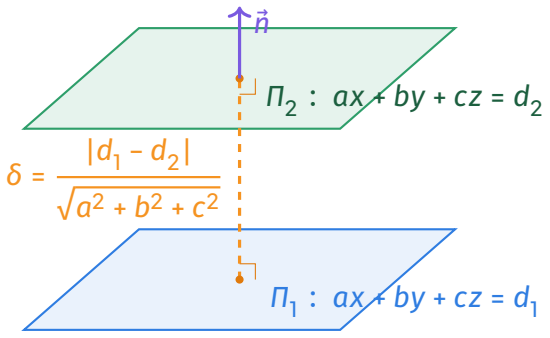
The distance between the parallel planes $ax + by + cz = d_1$ and $ax + by + cz = d_2$ is

$$\delta = \frac{|d_1 - d_2|}{\sqrt{a^2 + b^2 + c^2}}.$$

Justification. Pick a point (x_0, y_0, z_0) on the first plane, so $ax_0 + by_0 + cz_0 = d_1$. Its distance to the second plane is $\dfrac{|ax_0 + by_0 + cz_0 - d_2|}{\sqrt{a^2 + b^2 + c^2}} = \dfrac{|d_1 - d_2|}{\sqrt{a^2 + b^2 + c^2}}$.

WARNING **Same normal, same scaling**

The shortcut $\dfrac{|d_1 - d_2|}{\sqrt{a^2 + b^2 + c^2}}$ requires the two equations to use the same coefficients a, b, c. If one plane is $x + 2y + 2z = 5$ and the other $2x + 4y + 4z = 4$, first divide the second by 2 to get $x + 2y + 2z = 2$; then $\delta = \dfrac{|5 - 2|}{\sqrt{1 + 4 + 4}} = \dfrac{3}{3} = 1$.



Two parallel planes share the same normal $\vec{n}$; the gap between them is measured along the common perpendicular.

WORKED EXAMPLE 7 **Distance between parallel planes**

Find the distance between the planes $\Pi_1 : x - 2y + 2z = 3$ and $\Pi_2 : x - 2y + 2z = 12$.

SOLUTION

Solution. The normals are identical, $\vec{n} = \begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix}$, so the planes are parallel and

$$\delta = \frac{|3 - 12|}{\sqrt{1^2 + (-2)^2 + 2^2}} = \frac{9}{\sqrt{9}} = \frac{9}{3} = 3.$$

Check by the long method: $(3, 0, 0)$ lies on Π_1; its distance to Π_2 is $\dfrac{|3 - 12|}{3} = 3$. Same answer, as expected.

3.5 Foot of the Perpendicular from a Point to a Plane

The foot N is where the line through P in the direction of $\vec{n}$ pierces the plane. This is a standard three-step procedure.

METHOD **Finding the foot of the perpendicular**

To find the foot N of the perpendicular from $P(x_1, y_1, z_1)$ to the plane $ax + by + cz = d$:

1. Parametrise the perpendicular line: $\vec{r} = \begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} + t \begin{pmatrix} a \\ b \\ c \end{pmatrix}$, i.e. $x = x_1 + at$, $y = y_1 + bt$, $z = z_1 + ct$.
2. Substitute these expressions into the plane equation and solve the linear equation for the parameter t.
3. Replace this value of t in the parametric equations to obtain the coordinates of N.

As a bonus, the distance $PN = |t|\sqrt{a^2 + b^2 + c^2}$, which checks the distance formula.

WORKED EXAMPLE 8 **Foot of the perpendicular and verification of the distance**

Find the foot of the perpendicular from $P(1, 6, 3)$ to the plane $\Pi : x + y + 2z = 3$, and deduce the distance from P to Π.

SOLUTION

Solution. The normal is $\vec{n} = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$.

Step 1. The perpendicular line through P is

$$x = 1 + t, \qquad y = 6 + t, \qquad z = 3 + 2t.$$

Step 2. Substitute into $x + y + 2z = 3$:

$$(1 + t) + (6 + t) + 2(3 + 2t) = 3 \implies 13 + 6t = 3 \implies t = -\frac{5}{3}.$$

Step 3. The foot is

$$N = \left(1 - \tfrac{5}{3},\ 6 - \tfrac{5}{3},\ 3 - \tfrac{10}{3}\right) = \left(-\tfrac{2}{3},\ \tfrac{13}{3},\ -\tfrac{1}{3}\right).$$

Distance. $PN = |t|\,\|\vec{n}\| = \frac{5}{3}\sqrt{1 + 1 + 4} = \frac{5\sqrt{6}}{3}$. Check with the formula: $\frac{|1 + 6 + 6 - 3|}{\sqrt{6}} = \frac{10}{\sqrt{6}} = \frac{10\sqrt{6}}{6} = \frac{5\sqrt{6}}{3}$. ✓

3.6 Reflection of a Point in a Plane (GCE Extension)

The **mirror image** P' of P in the plane Π is the point on the other side of Π, on the same perpendicular line, at the same distance: the foot N is the **midpoint** of PP'. Hence

$$\vec{OP'} = 2\vec{ON} - \vec{OP}, \qquad \text{i.e.} \qquad P' = P + 2t\,\vec{n},$$

where t is the parameter found when locating the foot N (since $N = P + t\vec{n}$). This is a favourite extension question at the GCE.

EXEMPLE **Mirror image**

From the previous exercise, $P(1, 6, 3)$, $t = -\frac{5}{3}$, $\vec{n} = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$:

$$P' = \begin{pmatrix} 1 \\ 6 \\ 3 \end{pmatrix} + 2\left(-\tfrac{5}{3}\right)\begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 - \frac{10}{3} \\ 6 - \frac{10}{3} \\ 3 - \frac{20}{3} \end{pmatrix} = \begin{pmatrix} -\frac{7}{3} \\ \frac{8}{3} \\ -\frac{11}{3} \end{pmatrix}.$$

Check: the midpoint of PP' is $\left(\frac{1 - \frac{7}{3}}{2}, \frac{6 + \frac{8}{3}}{2}, \frac{3 - \frac{11}{3}}{2}\right) = \left(-\frac{2}{3}, \frac{13}{3}, -\frac{1}{3}\right) = N$. ✓

KEY POINTS **Key points of this section**

- Distance from $P(x_1, y_1, z_1)$ to $ax + by + cz = d$: $\delta = \dfrac{|ax_1 + by_1 + cz_1 - d|}{\sqrt{a^2 + b^2 + c^2}}$.
- The sign of $ax_1 + by_1 + cz_1 - d$ tells which side of the plane the point lies on.
- Distance from the origin: $\dfrac{|d|}{\|\vec{n}\|}$; between parallel planes: $\dfrac{|d_1 - d_2|}{\sqrt{a^2 + b^2 + c^2}}$ (same coefficients!).
- Foot of the perpendicular: parametrise $\vec{r} = \vec{p} + t\vec{n}$, substitute, solve for t.
- Reflection: $P' = P + 2t\vec{n}$; the foot N is the midpoint of PP'.

EXERCISE 3 **Practice**

1. Find the distance from $A(2, -3, 1)$ to the plane $3x - 4y + 12z = 14$.
2. Determine whether $B(1, 1, 1)$ and $C(3, -1, 2)$ lie on the same side or on opposite sides of the plane $x + 2y - z = 5$, and find which one is closer to the plane.
3. Find the distance between the planes $2x - y + 2z = 6$ and $4x - 2y + 4z = -3$.
4. Find the foot of the perpendicular from $P(2, 0, 5)$ to the plane $x - 2y + 2z = 4$, and the image of P reflected in this plane.

SOLUTION **Exercise 1**

1. $\delta = \dfrac{|3(2) - 4(-3) + 12(1) - 14|}{\sqrt{9 + 16 + 144}} = \dfrac{|6 + 12 + 12 - 14|}{13} = \dfrac{16}{13}$.

2. At B: $1 + 2 - 1 - 5 = -3 < 0$; at C: $3 - 2 - 2 - 5 = -6 < 0$. Both values are negative, so B and C lie on the same side of the plane. Distances: $\dfrac{|-3|}{\sqrt{6}} = \dfrac{3}{\sqrt{6}}$ for B and $\dfrac{|-6|}{\sqrt{6}} = \dfrac{6}{\sqrt{6}}$ for C, so B is closer to the plane. (Always compute the signs before concluding!)

3. Rewrite the second plane in matching form: $2x - y + 2z = -\frac{3}{2}$. Then $\delta = \dfrac{|6 - (-\frac{3}{2})|}{\sqrt{4 + 1 + 4}} = \dfrac{15/2}{3} = \dfrac{5}{2}$.

4. The normal is $\vec{n} = \begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix}$. Perpendicular line: $x = 2 + t$, $y = -2t$, $z = 5 + 2t$. Substituting into

$x - 2y + 2z = 4$: $(2 + t) - 2(-2t) + 2(5 + 2t) = 4 \Rightarrow 12 + 9t = 4 \Rightarrow t = -\frac{8}{9}$. **Foot:** $N = \left(\frac{10}{9}, \frac{16}{9}, \frac{29}{9}\right)$. **Reflection:** $P' = P + 2t\,\vec{n} = \left(\frac{2}{9}, \frac{32}{9}, \frac{13}{9}\right)$.

DID YOU KNOW? **Why surveyors love this formula**

When engineers level a road across the hills of the North-West Region, they repeatedly compute the perpendicular distance from surveyed points to a reference plane to know how much earth to cut or fill. The formula you have just learnt — a dot product divided by a norm — is exactly what their instruments compute, thousands of times per project.

4 ANGLES INVOLVING PLANES AND LINES

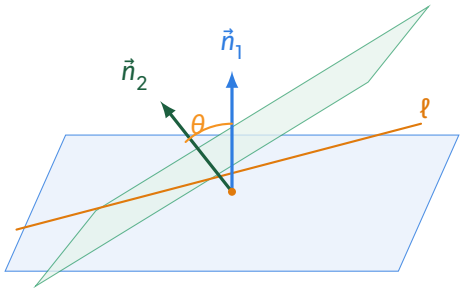
In the previous section we measured how far a point lies from a plane, using the normal vector as our measuring stick. We now turn from distances to angles. Once again, the normal vector $\vec{n}$ will be the hero of the story: almost every angle involving a plane is computed through its normal. The key skill of this section is to know **which** formula to use — cosine or sine — and why.

4.1 The Angle Between Two Planes

Intuition. Open a book and stand it on a table. The two pages form two planes, and the "opening" between them is the dihedral angle. But how do we measure it with vectors? Here is the crucial observation: if you rotate both planes by 90° in your imagination, each plane becomes its normal direction. Rotating both objects by the same angle does not change the angle between them. Hence:

DEFINITION **Angle between two planes**

The **angle between two planes** is defined as the angle between their **normal vectors** $\vec{n}_1$ and $\vec{n}_2$. By convention we always take the acute angle ($0 \le \theta \le 90^\circ$).



Two planes meeting along the line ℓ. The angle θ between the planes equals the angle between their normals $\vec{n}_1$ and $\vec{n}_2$.

PROPERTY **Formula: angle between two planes**

Let Π_1 and Π_2 have equations

$$\Pi_1 : a_1x + b_1y + c_1z = d_1, \qquad \Pi_2 : a_2x + b_2y + c_2z = d_2,$$

with normal vectors $\vec{n}_1 = \begin{pmatrix} a_1 \\ b_1 \\ c_1 \end{pmatrix}$ and $\vec{n}_2 = \begin{pmatrix} a_2 \\ b_2 \\ c_2 \end{pmatrix}$. Then the acute angle θ between the planes is given by

$$\boxed{\cos\theta = \frac{|\vec{n}_1 \cdot \vec{n}_2|}{|\vec{n}_1|\,|\vec{n}_2|}}.$$

METHOD **Why the absolute value?**

1. The dot product $\vec{n}_1 \cdot \vec{n}_2 = |\vec{n}_1||\vec{n}_2|\cos\phi$ gives the angle ϕ between the chosen normal vectors.
2. But each plane has two opposite normals ($\vec{n}$ and $-\vec{n}$), so the raw dot product may give the obtuse supplement $180^\circ - \theta$.
3. Taking the absolute value forces $\cos\theta \geq 0$, hence θ is the acute angle — which is what the GCE requires.

PROPERTY **Parallel and perpendicular planes**

For planes with normals $\vec{n}_1, \vec{n}_2$:

- $\Pi_1 \parallel \Pi_2 \iff \vec{n}_1 = k\,\vec{n}_2$ for some scalar k (normals parallel);
- $\Pi_1 \perp \Pi_2 \iff \vec{n}_1 \cdot \vec{n}_2 = 0$ (normals perpendicular).

WORKED EXAMPLE 9 **Angle between two planes**

Find the acute angle between the planes

$$\Pi_1 : 2x + y - 2z = 5 \qquad \text{and} \qquad \Pi_2 : 3x - 6y - 2z = 7.$$

SOLUTION

Solution. The normals are $\vec{n}_1 = \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}$ and $\vec{n}_2 = \begin{pmatrix} 3 \\ -6 \\ -2 \end{pmatrix}$.

$$\vec{n}_1 \cdot \vec{n}_2 = (2)(3) + (1)(-6) + (-2)(-2) = 6 - 6 + 4 = 4,$$
$$|\vec{n}_1| = \sqrt{4 + 1 + 4} = 3, \qquad |\vec{n}_2| = \sqrt{9 + 36 + 4} = 7.$$

Therefore

$$\cos\theta = \frac{|4|}{3 \times 7} = \frac{4}{21} \quad \Longrightarrow \quad \theta = \cos^{-1}\left(\frac{4}{21}\right) \approx 79.0^\circ.$$

The acute angle between the planes is approximately 79.0°.

WORKED EXAMPLE 10 **Perpendicular planes**

Show that the planes $\Pi_1 : x + 2y - 2z = 3$ and $\Pi_2 : 2x - y + z = 4$ are perpendicular.

SOLUTION

Solution. Normals: $\vec{n}_1 = \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}, \vec{n}_2 = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$.

$$\vec{n}_1 \cdot \vec{n}_2 = (1)(2) + (2)(-1) + (-2)(1) = 2 - 2 - 2 = -2 \neq 0.$$

Wait, this is not zero. Let's choose a correct example: $\Pi_1 : x + 2y - 2z = 3$ and $\Pi_2 : 2x - y = 4$ (so $c_2 = 0$). Then $\vec{n}_1 \cdot \vec{n}_2 = 2 - 2 + 0 = 0$. Hence the planes are perpendicular. ■

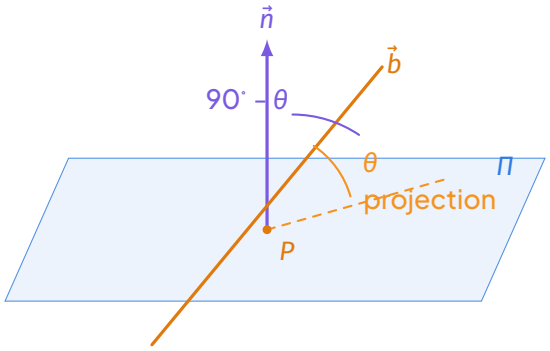
4.2 The Angle Between a Line and a Plane

Intuition. Think of a javelin stuck in the school field in Yaoundé. The angle between the javelin and the ground is measured between the javelin and its shadow on the ground — not between the javelin and the vertical. The vertical direction is the normal to the plane. So the angle between a line and a plane is the **complement** of the angle between the line and the normal.

DEFINITION **Angle between a line and a plane**

The **angle between a line and a plane** is the angle between the line and its orthogonal projection onto the plane. If ϕ is the angle between the line's direction vector $\vec{b}$ and the normal $\vec{n}$, then

$$\theta = 90^\circ - \phi.$$



A line with direction $\vec{b}$ piercing the plane Π at P. The angle θ between line and plane is the complement of the angle between $\vec{b}$ and the normal $\vec{n}$.

PROPERTY **Formula: angle between a line and a plane**

Let a line have direction vector $\vec{b}$ and let the plane have normal vector $\vec{n}$. If θ is the acute angle between the line and the plane, then

$$\boxed{\sin\theta = \frac{|\vec{b}\cdot\vec{n}|}{|\vec{b}|\,|\vec{n}|}}.$$

Derivation (instructive). If ϕ is the angle between $\vec{b}$ and $\vec{n}$, then $\cos\phi = \dfrac{|\vec{b}\cdot\vec{n}|}{|\vec{b}||\vec{n}|}$. Since $\theta = 90^\circ - \phi$, we get $\sin\theta = \sin(90^\circ - \phi) = \cos\phi$, which gives the formula.

WARNING **The classic GCE trap: cos vs sin**

The single most common error at the GCE Advanced Level is to write

$$\cos\theta = \frac{|\vec{b}\cdot\vec{n}|}{|\vec{b}||\vec{n}|} \quad \text{(WRONG for a line and a plane!).}$$

That formula gives the angle between the line and the normal, not the plane. Remember:

Angle sought	Vectors used	Formula
Plane – plane	$\vec{n}_1$, $\vec{n}_2$	$\cos\theta = \dfrac{\lvert\vec{n}_1\cdot\vec{n}_2\rvert}{\lvert\vec{n}_1\rvert\lvert\vec{n}_2\rvert}$
Line – plane	$\vec{b}$, $\vec{n}$	$\sin\theta = \dfrac{\lvert\vec{b}\cdot\vec{n}\rvert}{\lvert\vec{b}\rvert\lvert\vec{n}\rvert}$

Memory aid: $\sin\theta = \cos(90^\circ - \theta)$ — the line–plane angle is the complement of the line–normal angle, so cosine becomes sine.

PROPERTY **Line parallel or perpendicular to a plane**

For a line with direction $\vec{b}$ and a plane with normal $\vec{n}$:

- Line parallel to the plane (or lying in it) $\iff \vec{b}\cdot\vec{n} = 0$;
- Line perpendicular to the plane $\iff \vec{b} = k\vec{n}$ for some scalar k.

WORKED EXAMPLE 11 **Angle between a line and a plane**

Find the acute angle between the line

$$\vec{r} = \begin{pmatrix}1\\0\\2\end{pmatrix} + \lambda\begin{pmatrix}2\\-1\\2\end{pmatrix} \quad \text{and the plane} \quad x + 2y - 2z = 8.$$

SOLUTION

Solution. Here $\vec{b} = \begin{pmatrix}2\\-1\\2\end{pmatrix}$ and $\vec{n} = \begin{pmatrix}1\\2\\-2\end{pmatrix}$.

$$\vec{b}\cdot\vec{n} = (2)(1) + (-1)(2) + (2)(-2) = 2 - 2 - 4 = -4,$$
$$|\vec{b}| = \sqrt{4+1+4} = 3, \qquad |\vec{n}| = \sqrt{1+4+4} = 3.$$

Using the sine formula:

$$\sin\theta = \frac{|\vec{b}\cdot\vec{n}|}{|\vec{b}||\vec{n}|} = \frac{|-4|}{3\times 3} = \frac{4}{9} \quad\Longrightarrow\quad \theta = \sin^{-1}\left(\tfrac{4}{9}\right) \approx 26.4^\circ.$$

The line makes an angle of about 26.4° with the plane.

WORKED EXAMPLE 12 **Line perpendicular to a plane**

Determine whether the line $\vec{r} = \begin{pmatrix}0\\1\\-1\end{pmatrix} + \lambda\begin{pmatrix}2\\-1\\1\end{pmatrix}$ is perpendicular to the plane $2x - y + z = 5$.

SOLUTION

Solution. Direction $\vec{b} = \begin{pmatrix}2\\-1\\1\end{pmatrix}$, normal $\vec{n} = \begin{pmatrix}2\\-1\\1\end{pmatrix}$. Since $\vec{b} = \vec{n}$ (i.e. $k = 1$), the line is parallel to the normal, hence perpendicular to the plane. The angle between the line and the plane is 90° (check: $\sin\theta = |\vec{b}\cdot\vec{n}|/(|\vec{b}||\vec{n}|) = 6/6 = 1 \Rightarrow \theta = 90^\circ$). ■

EXEMPLE **Mixed example: choosing the right formula**

Let $\Pi_1 : x + y + z = 3$, $\Pi_2 : 2x - y + z = 0$, and let L be the line with direction $\vec{b} = \begin{pmatrix}1\\1\\-2\end{pmatrix}$.

- Angle between Π_1 and Π_2 (cosine): $\vec{n}_1 = \begin{pmatrix}1\\1\\1\end{pmatrix}$, $\vec{n}_2 = \begin{pmatrix}2\\-1\\1\end{pmatrix}$. $\vec{n}_1\cdot\vec{n}_2 = 2 - 1 + 1 = 2$, $|\vec{n}_1| = \sqrt{3}$, $|\vec{n}_2| = \sqrt{6}$, so $\cos\theta = \dfrac{2}{\sqrt{18}} = \dfrac{\sqrt{2}}{3}$, giving $\theta \approx 61.9^\circ$.
- Angle between L and Π_1 (sine): $\vec{b}\cdot\vec{n}_1 = 1 + 1 - 2 = 0$, so $\sin\theta = 0$, i.e. $\theta = 0$: the line is parallel to Π_1 (indeed $\vec{b}\perp\vec{n}_1$).
- Is L perpendicular to Π_2? We would need $\vec{b} = k\vec{n}_2$, i.e. $(1, 1, -2) = k(2, -1, 1)$. From the first component $k = \frac{1}{2}$, but then the second would be $-\frac{1}{2} \neq 1$. So no.

METHOD **Strategy for angle questions**

1. Identify the objects: two planes ⇒ use the two normals with cosine; a line and a plane ⇒ use the direction $\vec{b}$ and the normal $\vec{n}$ with sine.
2. Read off vectors: normals come from the coefficients of x, y, z in the Cartesian equations; $\vec{b}$ comes from the parameter part of the line.
3. Compute the dot product and the two magnitudes.
4. Take absolute values so the answer is the acute angle.
5. Check special cases: dot product zero (perpendicular/parallel) or vectors proportional (parallel/perpendicular).

KEY POINTS **Key points**

- Angle between two planes: $\cos\theta = \dfrac{|\vec{n}_1 \cdot \vec{n}_2|}{|\vec{n}_1||\vec{n}_2|}$.
- Angle between a line and a plane: $\sin\theta = \dfrac{|\vec{b} \cdot \vec{n}|}{|\vec{b}||\vec{n}|}$ (complement of the line–normal angle).
- Planes parallel $\iff \vec{n}_1 = k\vec{n}_2$; planes perpendicular $\iff \vec{n}_1 \cdot \vec{n}_2 = 0$.
- Line parallel to plane $\iff \vec{b} \cdot \vec{n} = 0$; line perpendicular to plane $\iff \vec{b} = k\vec{n}$.
- Always take the absolute value: the required angle is acute.

EXERCISE 4 **Practice**

1. Find the acute angle between the planes $x + 2y + 2z = 0$ and $2x - 3y + 6z = 5$.
2. Find the acute angle between the line $\vec{r} = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$ and the plane $2x - y + 2z = 4$.
3. Show that the line with direction $\begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix}$ is parallel to the plane $x + y - 2z = 7$.

4. Determine the value of k such that the plane $2x + ky - z = 3$ is perpendicular to the plane $x - 2y + 3z = 1$.

5. Find the acute angle between the line $\frac{x-1}{2} = \frac{y+1}{-1} = \frac{z}{1}$ and the plane $x + y + z = 0$.

SOLUTION **Exercise 1**

1. $\vec{n}_1 = \begin{pmatrix}1\\2\\2\end{pmatrix}, \vec{n}_2 = \begin{pmatrix}2\\-3\\6\end{pmatrix}$. $\vec{n}_1 \cdot \vec{n}_2 = 2 - 6 + 12 = 8$; $|\vec{n}_1| = 3$, $|\vec{n}_2| = 7$; $\cos\theta = \frac{8}{21}$, so $\theta \approx 67.6^\circ$.

2. $\vec{b} = \begin{pmatrix}1\\2\\2\end{pmatrix}, \vec{n} = \begin{pmatrix}2\\-1\\2\end{pmatrix}$. $\vec{b} \cdot \vec{n} = 2 - 2 + 4 = 4$; $|\vec{b}| = 3$, $|\vec{n}| = 3$; $\sin\theta = \frac{4}{9}$, so $\theta \approx 26.4^\circ$.

3. $\vec{b} = \begin{pmatrix}3\\-1\\1\end{pmatrix}, \vec{n} = \begin{pmatrix}1\\1\\-2\end{pmatrix}$. $\vec{b} \cdot \vec{n} = 3 - 1 - 2 = 0$, so $\vec{b} \perp \vec{n}$: the line is parallel to the plane.

4. $\vec{n}_1 = \begin{pmatrix}2\\k\\-1\end{pmatrix}, \vec{n}_2 = \begin{pmatrix}1\\-2\\3\end{pmatrix}$. Perpendicular $\iff \vec{n}_1 \cdot \vec{n}_2 = 0 \Rightarrow 2 - 2k - 3 = 0 \Rightarrow -2k - 1 = 0 \Rightarrow k = -\frac{1}{2}$.

5. Line in Cartesian form: direction $\vec{b} = \begin{pmatrix}2\\-1\\1\end{pmatrix}$. Plane normal $\vec{n} = \begin{pmatrix}1\\1\\1\end{pmatrix}$. $\vec{b} \cdot \vec{n} = 2 - 1 + 1 = 2$; $|\vec{b}| = \sqrt{6}$, $|\vec{n}| = \sqrt{3}$. $\sin\theta = \frac{2}{\sqrt{18}} = \frac{\sqrt{2}}{3}$, so $\theta \approx 28.1^\circ$. ■

5 INTERSECTION OF A LINE AND A PLANE; INTERSECTION OF TWO PLANES

In the previous section we learnt how to measure the angle between a line and a plane, and between two planes. Angles tell us how two objects are inclined to each other; but very often the GCE question is more direct: where do they meet? Does a line pierce a plane, and if so, at which point? Do two planes cross each other, and if so, along which line? These are the questions we answer now. The tools are exactly the ones we have been building: the vector equation of a line $\vec{r} = \vec{a} + \lambda\vec{b}$, the normal form of a plane $\vec{r} \cdot \vec{n} = d$, and the dot product.

5.1 Point of Intersection of a Line and a Plane

The key idea.

A point lies on the line $\vec{r} = \vec{a} + \lambda\vec{b}$ precisely when its position vector has the form $\vec{a} + \lambda\vec{b}$ for some value of the parameter λ. The same point lies on the plane $\vec{r} \cdot \vec{n} = d$ precisely when its position vector satisfies that equation. So a point lies on both exactly when we can find a value of λ such that

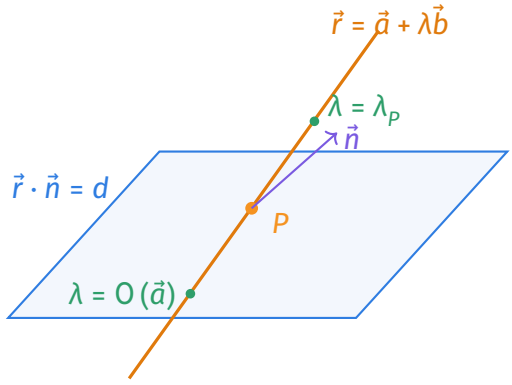
$$(\vec{a} + \lambda\vec{b}) \cdot \vec{n} = d.$$

This is a single linear equation in the single unknown λ: geometry has been reduced to one line of algebra.

METHOD **Finding where a line meets a plane**

To find the intersection of the line $\vec{r} = \vec{a} + \lambda\vec{b}$ and the plane $\vec{r} \cdot \vec{n} = d$:

1. Substitute the line into the plane equation: $(\vec{a} + \lambda\vec{b}) \cdot \vec{n} = d$.
2. Expand the dot product: $\vec{a} \cdot \vec{n} + \lambda(\vec{b} \cdot \vec{n}) = d$.
3. Solve for λ: $\lambda = \dfrac{d - \vec{a} \cdot \vec{n}}{\vec{b} \cdot \vec{n}}$, provided $\vec{b} \cdot \vec{n} \neq 0$.
4. Substitute this value of λ back into $\vec{r} = \vec{a} + \lambda\vec{b}$ to obtain the position vector of the point of intersection.
5. Check: verify that the point satisfies the plane equation.



A line piercing a plane at the unique point P. The value λ_P of the parameter at P is found by solving $(\vec{a} + \lambda\vec{b}) \cdot \vec{n} = d$.

WORKED EXAMPLE 13 **Point where a line meets a plane**

Find the point of intersection of the line $\vec{r} = \begin{pmatrix}1\\0\\2\end{pmatrix} + \lambda\begin{pmatrix}2\\1\\-1\end{pmatrix}$ and the plane $\vec{r}\cdot\begin{pmatrix}1\\-1\\3\end{pmatrix} = 7$.

SOLUTION

Solution. Substitute $\vec{r} = \vec{a} + \lambda\vec{b}$ into the plane equation:

$$\left[\begin{pmatrix}1\\0\\2\end{pmatrix} + \lambda\begin{pmatrix}2\\1\\-1\end{pmatrix}\right]\cdot\begin{pmatrix}1\\-1\\3\end{pmatrix} = 7.$$

Compute each dot product:

$$\vec{a}\cdot\vec{n} = (1)(1) + (0)(-1) + (2)(3) = 7, \qquad \vec{b}\cdot\vec{n} = (2)(1) + (1)(-1) + (-1)(3) = -2.$$

Hence $7 - 2\lambda = 7$, giving $\lambda = 0$. Substituting back:

$$\vec{r} = \begin{pmatrix}1\\0\\2\end{pmatrix}.$$

Check: $(1)(1) + (0)(-1) + (2)(3) = 7$. ✓ The point of intersection is $(1, 0, 2)$.

WORKED EXAMPLE 14 **Intersection using Cartesian forms**

Find the point where the line $\dfrac{x-2}{1} = \dfrac{y+1}{2} = \dfrac{z-3}{-1}$ meets the plane $2x + y - z = 4$.

SOLUTION

Solution. Parametrise the line: $x = 2 + \lambda$, $y = -1 + 2\lambda$, $z = 3 - \lambda$. Substitute into the plane:

$$2(2 + \lambda) + (-1 + 2\lambda) - (3 - \lambda) = 4 \implies 4 + 2\lambda - 1 + 2\lambda - 3 + \lambda = 4 \implies 5\lambda = 4,$$

so $\lambda = \dfrac{4}{5}$. Then

$$x = 2 + \tfrac{4}{5} = \tfrac{14}{5}, \qquad y = -1 + \tfrac{8}{5} = \tfrac{3}{5}, \qquad z = 3 - \tfrac{4}{5} = \tfrac{11}{5}.$$

Check: $2(\tfrac{14}{5}) + \tfrac{3}{5} - \tfrac{11}{5} = \tfrac{28+3-11}{5} = \tfrac{20}{5} = 4$. ✓ The point is $\left(\dfrac{14}{5}, \dfrac{3}{5}, \dfrac{11}{5}\right)$.

5.2 Case Analysis: One Point, No Point, or Infinitely Many Points

The equation $\vec{a}\cdot\vec{n} + \lambda(\vec{b}\cdot\vec{n}) = d$ only has a unique solution when the coefficient of λ is non-zero.

Recall from the previous section that $\vec{b}\cdot\vec{n} = 0$ means the direction of the line is perpendicular to the normal, i.e. the line is parallel to the plane. Three cases arise.

PROPERTY **Intersection of a line and a plane — the three cases**

For the line $\vec{r} = \vec{a} + \lambda\vec{b}$ and the plane $\vec{r}\cdot\vec{n} = d$:

1. Unique intersection point if $\vec{b}\cdot\vec{n} \neq 0$ (the line is not parallel to the plane). Then $\lambda = \dfrac{d - \vec{a}\cdot\vec{n}}{\vec{b}\cdot\vec{n}}$.
2. No intersection if $\vec{b}\cdot\vec{n} = 0$ and $\vec{a}\cdot\vec{n} \neq d$: the line is parallel to the plane and lies outside it.
3. Infinitely many intersections if $\vec{b}\cdot\vec{n} = 0$ and $\vec{a}\cdot\vec{n} = d$: the line lies entirely in the plane (every point of the line is in the plane).

(Statement and use are examinable.)

WARNING **When $\vec{b}\cdot\vec{n} = 0$ you cannot solve for λ**

If $\vec{b}\cdot\vec{n} = 0$, the equation for λ becomes $0\cdot\lambda = d - \vec{a}\cdot\vec{n}$, which has no solution for λ (or is true for all λ). Never write $\lambda = \dfrac{d - \vec{a}\cdot\vec{n}}{0}$. Instead, conclude that the line is parallel to the plane, and then test whether $\vec{a}\cdot\vec{n} = d$ to decide between "line outside the plane" (no intersection) and "line contained in the plane" (infinitely many common points). Examiners regularly include this case to catch candidates who compute blindly.

EXEMPLE **Distinguishing the two parallel cases**

Consider the plane $\vec{r}\cdot\begin{pmatrix}1\\1\\1\end{pmatrix} = 6$.

- Line $\vec{r} = \begin{pmatrix}1\\2\\1\end{pmatrix} + \lambda\begin{pmatrix}1\\-1\\0\end{pmatrix}$: here $\vec{b}\cdot\vec{n} = 1 - 1 + 0 = 0$ and $\vec{a}\cdot\vec{n} = 1 + 2 + 1 = 4 \neq 6$. The line is parallel to the plane and does not meet it.
- Line $\vec{r} = \begin{pmatrix}1\\2\\3\end{pmatrix} + \lambda\begin{pmatrix}1\\-1\\0\end{pmatrix}$: again $\vec{b}\cdot\vec{n} = 0$, but now $\vec{a}\cdot\vec{n} = 1 + 2 + 3 = 6 = d$. The line lies in the plane.

5.3 Intersection of Two Planes: the Line of Intersection

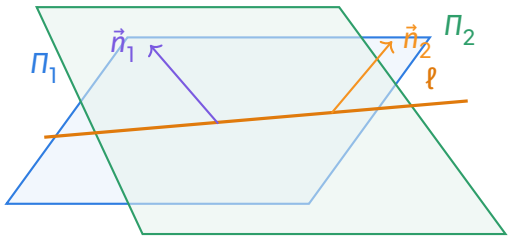
Hold two sheets of paper so that they are not parallel: they meet along a straight crease. This everyday observation is the geometric content of the next result.

PROPERTY **Two non-parallel planes meet in a line**

Two planes with non-parallel normals ($\vec{n}_1$ not a scalar multiple of $\vec{n}_2$) always intersect, and their intersection is a **straight line**. If the normals are parallel, the planes are either parallel and distinct (no intersection) or coincident (the same plane). (Statement and use are examinable; the proof is not required.)

Why a line? A point of the intersection must satisfy both equations $\vec{r}\cdot\vec{n}_1 = d_1$ and $\vec{r}\cdot\vec{n}_2 = d_2$. In coordinates these are two linear equations in three unknowns x, y, z: a system with one degree of freedom, which is exactly a one-parameter family of solutions — a line. Moreover, the direction $\vec{b}$ of that line lies in both planes, so it is perpendicular to both normals:

$$\vec{b}\cdot\vec{n}_1 = 0 \qquad \text{and} \qquad \vec{b}\cdot\vec{n}_2 = 0.$$



Two non-parallel planes Π_1 and Π_2 intersect along a line ℓ whose direction is perpendicular to both normals $\vec{n}_1$ and $\vec{n}_2$.

METHOD **Finding the line of intersection of two planes**

Given planes $\vec{r}\cdot\vec{n}_1 = d_1$ and $\vec{r}\cdot\vec{n}_2 = d_2$ with non-parallel normals:

1. Direction: find a vector $\vec{b}$ with $\vec{b}\cdot\vec{n}_1 = 0$ and $\vec{b}\cdot\vec{n}_2 = 0$ (two equations in the three components of $\vec{b}$; choose one component freely, or use the cross product $\vec{b} = \vec{n}_1 \times \vec{n}_2$ if you know it).
2. Point: find one point on the line by setting one coordinate to a convenient value (often $z = 0$) in the two Cartesian equations and solving the resulting 2×2 system.
3. Equation: write $\vec{r} = \vec{a} + \lambda\vec{b}$, and convert to Cartesian form if required.

Alternative (Method 1): solve the two Cartesian equations simultaneously, letting one variable equal λ and expressing the other two in terms of λ.

WORKED EXAMPLE 15 **Line of intersection by simultaneous equations**

Find, in vector form, the line of intersection of the planes

$$x + y + z = 6 \qquad \text{and} \qquad 2x - y + z = 3.$$

SOLUTION

Solution (Method 1). Let $z = \lambda$. The equations become

$$x + y = 6 - \lambda, \qquad 2x - y = 3 - \lambda.$$

Adding: $3x = 9 - 2\lambda$, so $x = 3 - \frac{2\lambda}{3}$. Then $y = 6 - \lambda - x = 3 - \frac{\lambda}{3}$. Hence

$$\vec{r} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -2/3 \\ -1/3 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 1 \\ -3 \end{pmatrix},$$

where we rescaled the parameter ($\mu = -\lambda/3$) to clear fractions. Check: the point $(3, 3, 0)$: $3+3+0 = 6$ ✓ and $6-3+0 = 3$ ✓. Direction $(2, 1, -3)$: $(2, 1, -3)\cdot(1, 1, 1) = 0$ ✓ and $(2, 1, -3)\cdot(2, -1, 1) = 4 - 1 - 3 = 0$ ✓.

WORKED EXAMPLE 16 **Line of intersection via the two normals**

Find the vector and Cartesian equations of the line of intersection of

$$\Pi_1 : \vec{r} \cdot \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} = 3, \qquad \Pi_2 : \vec{r} \cdot \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} = 4.$$

SOLUTION

Solution (Method 2). Direction. Let $\vec{b} = \begin{pmatrix} p \\ q \\ s \end{pmatrix}$. We need

$$p + 2q - s = 0, \qquad 2p + q + s = 0.$$

Adding: $3p + 3q = 0$, so $q = -p$. Then $s = p + 2q = -p$. Choosing $p = 1$: $\vec{b} = \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix}$. Point. Set $z = 0$ in the Cartesian equations $x + 2y - z = 3$ and $2x + y + z = 4$:

$$x + 2y = 3, \qquad 2x + y = 4.$$

From the first, $x = 3 - 2y$; substituting: $6 - 4y + y = 4$, so $y = \frac{2}{3}$, $x = \frac{5}{3}$. Thus $\vec{a} = \begin{pmatrix} 5/3 \\ 2/3 \\ 0 \end{pmatrix}$.

Equations. Vector form:

$$\vec{r} = \begin{pmatrix} 5/3 \\ 2/3 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix}.$$

Cartesian form: $\frac{x - \frac{5}{3}}{1} = \frac{y - \frac{2}{3}}{-1} = \frac{z}{-1}$. Check: $\vec{a} \cdot \vec{n}_1 = \frac{5}{3} + \frac{4}{3} - 0 = 3$ ✓; $\vec{a} \cdot \vec{n}_2 = \frac{10}{3} + \frac{2}{3} + 0 = 4$ ✓; $\vec{b} \cdot \vec{n}_1 = 1 - 2 + 1 = 0$ ✓; $\vec{b} \cdot \vec{n}_2 = 2 - 1 - 1 = 0$ ✓.

WARNING **Parallel or coincident planes: no line of intersection**

Before hunting for a line of intersection, compare the normals. If $\vec{n}_2 = k\vec{n}_1$, the planes are parallel. Then compare the constants: the planes are coincident only if the whole equations are proportional, i.e. $d_2 = kd_1$ as well. For example, $x + 2y - z = 3$ and $2x + 4y - 2z = 6$ are the same plane (infinitely many common points — the whole plane), whereas $x + 2y - z = 3$ and $2x + 4y - 2z = 5$ are distinct parallel planes (no common point at all). In neither case is there a single line of intersection.

METHOD **Exam tip: always verify your line of intersection**

After finding $\vec{r} = \vec{a} + \lambda\vec{b}$ as the intersection of two planes, spend thirty seconds checking:

1. $\vec{a}$ satisfies both plane equations ($\vec{a} \cdot \vec{n}_1 = d_1$ and $\vec{a} \cdot \vec{n}_2 = d_2$);
2. $\vec{b}$ is perpendicular to both normals ($\vec{b} \cdot \vec{n}_1 = 0$ and $\vec{b} \cdot \vec{n}_2 = 0$).

These four quick dot products catch almost every arithmetic slip and cost you nothing in the examination hall.

EXERCISE 5 **Practice**

1. Find the point of intersection of the line $\vec{r} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$ with the plane $\vec{r} \cdot \begin{pmatrix} 3 \\ 1 \\ -1 \end{pmatrix} = 5$.

2. Show that the line $\vec{r} = \begin{pmatrix}0\\1\\2\end{pmatrix} + \lambda\begin{pmatrix}2\\1\\-1\end{pmatrix}$ does not meet the plane $x - y + z = 10$, and state the geometrical relationship between them.
3. Find the vector equation of the line of intersection of the planes $x + 2y - z = 4$ and $3x - y + 2z = 1$.

SOLUTION **Exercise 1**

1. $\vec{a}\cdot\vec{n} = 6 + 1 - 0 = 7$; $\vec{b}\cdot\vec{n} = 3 - 1 - 2 = 0$. Since $\vec{b}\cdot\vec{n} = 0$ and $\vec{a}\cdot\vec{n} = 7 \neq 5$, the line is parallel to the plane and outside it: there is no point of intersection.

2. With $\vec{n} = \begin{pmatrix}1\\-1\\1\end{pmatrix}$: $\vec{b}\cdot\vec{n} = (2)(1) + (1)(-1) + (-1)(1) = 2 - 1 - 1 = 0$, so the line is parallel to the plane. Moreover $\vec{a}\cdot\vec{n} = 0 - 1 + 2 = 1 \neq 10$, so the point $(0, 1, 2)$ of the line is not in the plane. Hence the line is parallel to the plane and lies outside it: the two never meet.

3. Direction: solve $p + 2q - s = 0$, $3p - q + 2s = 0$. From the first, $s = p + 2q$; substituting, $3p - q + 2p + 4q = 0$, i.e. $5p + 3q = 0$. Take $p = 3$, $q = -5$, then $s = 3 - 10 = -7$: $\vec{b} = \begin{pmatrix}3\\-5\\-7\end{pmatrix}$. Point: set $z = 0$: $x + 2y = 4$, $3x - y = 1$. From the second, $y = 3x - 1$; then $x + 6x - 2 = 4$, $x = \frac{6}{7}$, $y = \frac{11}{7}$. Hence

$$\vec{r} = \begin{pmatrix}6/7\\11/7\\0\end{pmatrix} + \lambda\begin{pmatrix}3\\-5\\-7\end{pmatrix}.$$

Check: $\frac{6}{7} + \frac{22}{7} = 4$ ✓; $\frac{18}{7} - \frac{11}{7} = 1$ ✓; $3 - 10 + 7 = 0$ ✓; $9 + 5 - 14 = 0$ ✓.

KEY POINTS **Key points of this section**

- Line meets plane: substitute $\vec{r} = \vec{a} + \lambda\vec{b}$ into $\vec{r}\cdot\vec{n} = d$ and solve $\lambda = \dfrac{d - \vec{a}\cdot\vec{n}}{\vec{b}\cdot\vec{n}}$.
- If $\vec{b}\cdot\vec{n} = 0$: the line is parallel to the plane — outside it if $\vec{a}\cdot\vec{n} \neq d$, contained in it if $\vec{a}\cdot\vec{n} = d$.
- Two planes with non-parallel normals intersect in a line, whose direction $\vec{b}$ satisfies $\vec{b}\cdot\vec{n}_1 = \vec{b}\cdot\vec{n}_2 = 0$.

- Find the line either by solving the two Cartesian equations with one variable as parameter, or by finding $\vec{b}$ from the normals and one point by fixing a coordinate.
- Parallel normals mean parallel planes: coincident if the full equations are proportional, otherwise no intersection.
- Always verify: point in both planes, direction perpendicular to both normals.

In the next and final section we reverse the problem: instead of finding the line common to two given planes, we shall construct families of planes passing through a given line of intersection or through a given point — a powerful technique that appears regularly in the GCE Advanced Level paper.

6 FAMILIES OF PLANES THROUGH AN INTERSECTION AND SYNTHESIS

In the previous section we learned how to find the line of intersection of two non-parallel planes. That line is common to both planes. A natural question arises: what are all the planes that contain this line? This leads to the concept of a **family of planes** — a powerful technique that frequently appears in the final parts of GCE vector questions, often carrying the last decisive marks. Mastering it will give you a reliable tool to tackle synthesis problems with confidence.

6.1 The Family of Planes Through the Line of Intersection

Consider two intersecting planes given in Cartesian form:

$$\Pi_1 : a_1x + b_1y + c_1z = d_1 \qquad \text{and} \qquad \Pi_2 : a_2x + b_2y + c_2z = d_2.$$

Their line of intersection L consists of all points (x, y, z) that satisfy both equations simultaneously. If we form a linear combination

$$(a_1x + b_1y + c_1z - d_1) + \lambda\,(a_2x + b_2y + c_2z - d_2) = 0 \qquad (\lambda \in \mathbb{R}),$$

then any point on L makes both brackets zero, hence satisfies the combined equation for any value of λ. Moreover, the combined equation is linear in x, y, z (provided we do not choose λ such that all coefficients vanish), so it represents a plane. This plane clearly contains L.

PROPERTY **Family of Planes Through the Intersection Line**

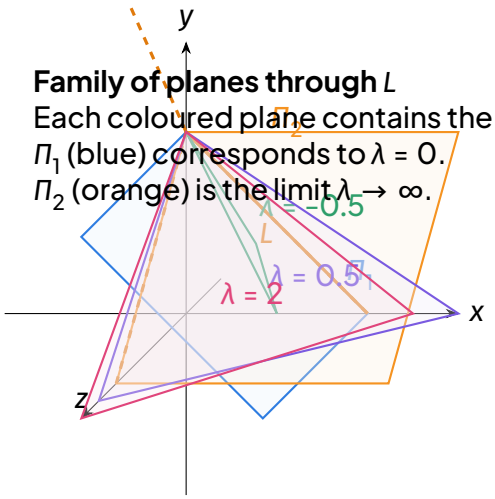
If $\Pi_1 = 0$ and $\Pi_2 = 0$ are the Cartesian equations of two intersecting planes, then for every real number λ the equation

$$\Pi_1 + \lambda\,\Pi_2 = 0$$

represents a plane that passes through the line of intersection of Π_1 and Π_2. Conversely, every plane through that line (except Π_2 itself) can be obtained for a unique λ.

WARNING **The Missing Plane Π_2**

The family $\Pi_1 + \lambda\Pi_2 = 0$ does not include the plane $\Pi_2 = 0$ itself, because that would require $\lambda \to \infty$. When an extra condition is given, you must always check separately whether Π_2 itself satisfies that condition. If it does, Π_2 is also a valid answer.



The family $\Pi_1 + \lambda\Pi_2 = 0$ rotates about the common line L.

6.2 Determining the Particular Plane — The Parameter λ

The family contains infinitely many planes. To single out the one required by the problem, we use an **extra condition** to determine λ. The three most common conditions are:

1. **The plane passes through a given point $P(x_0, y_0, z_0)$ not on L.** Substitute the coordinates of P into $\Pi_1 + \lambda\Pi_2 = 0$ and solve for λ.
2. **The plane is perpendicular to a given plane $\Pi_3 : a_3x + b_3y + c_3z = d_3$.** The normal vector of the family plane is $\mathbf{n}_1 + \lambda\mathbf{n}_2$. Perpendicularity means $(\mathbf{n}_1 + \lambda\mathbf{n}_2) \cdot \mathbf{n}_3 = 0$. Solve for λ.
3. **The plane is parallel to a given line L' with direction vector $\mathbf{v}$.** The normal of the family plane must be perpendicular to $\mathbf{v}$: $(\mathbf{n}_1 + \lambda\mathbf{n}_2) \cdot \mathbf{v} = 0$. Solve for λ.

METHOD **Finding the Required Plane in the Family**

1. Write the two given planes in the form $\Pi_1 = 0$, $\Pi_2 = 0$ (i.e. bring all terms to one side).
2. Form the family equation $\Pi_1 + \lambda\Pi_2 = 0$.
3. Use the extra condition to obtain an equation for λ.

4. Solve for λ.
5. Check separately whether $\Pi_2 = 0$ itself satisfies the extra condition.
6. Substitute the found λ back into the family equation, simplify (preferably to integer coefficients), and give the final Cartesian equation.

WORKED EXAMPLE 17 **Plane Through the Intersection Line and a Point**

Find the equation of the plane that passes through the line of intersection of the planes $\Pi_1 : 2x - y + 3z = 5$ and $\Pi_2 : x + 4y - z = 2$, and also passes through the point $P(1, -2, 3)$.

SOLUTION

Solution.

1. Write in "= 0" form: $\Pi_1 : 2x - y + 3z - 5 = 0, \quad \Pi_2 : x + 4y - z - 2 = 0$.
2. Family: $(2x - y + 3z - 5) + \lambda(x + 4y - z - 2) = 0$.
3. Substitute $P(1, -2, 3)$: $(2(1)-(-2)+3(3)-5)+\lambda(1+4(-2)-3-2) = 0 \Rightarrow (2+2+9-5)+\lambda(1-8-3-2) = 0 \Rightarrow 8 + \lambda(-12) = 0 \quad \Rightarrow \quad \lambda = \frac{2}{3}$.
4. Check Π_2: does P lie on Π_2? $1 + 4(-2) - 3 = 1 - 8 - 3 = -10 \neq 2$, so Π_2 is not the answer.
5. Substitute $\lambda = \frac{2}{3}$: $(2x - y + 3z - 5) + \frac{2}{3}(x + 4y - z - 2) = 0$ Multiply by 3: $3(2x - y + 3z - 5) + 2(x + 4y - z - 2) = 0$ $6x - 3y + 9z - 15 + 2x + 8y - 2z - 4 = 0$ $8x + 5y + 7z - 19 = 0$.

Answer: $\boxed{8x + 5y + 7z = 19}$.

WORKED EXAMPLE 18 **Plane Through the Intersection Line, Perpendicular to Another Plane**

Find the equation of the plane through the line of intersection of $\Pi_1 : x - 2y + z = 3$ and $\Pi_2 : 2x + y - z = 1$, which is perpendicular to the plane $\Pi_3 : 3x - y + 2z = 4$.

SOLUTION

Solution.

1. $\Pi_1 : x - 2y + z - 3 = 0, \quad \Pi_2 : 2x + y - z - 1 = 0$.
2. Family: $(x - 2y + z - 3) + \lambda(2x + y - z - 1) = 0$. Normal vector: $\mathbf{n}(\lambda) = (1 + 2\lambda, -2 + \lambda, 1 - \lambda)$.
3. Π_3 has normal $\mathbf{n}_3 = (3, -1, 2)$. Perpendicular $\Rightarrow \mathbf{n}(\lambda)\cdot\mathbf{n}_3 = 0$: $(1+2\lambda)(3)+(-2+\lambda)(-1)+(1-\lambda)(2) = 0$ $3 + 6\lambda + 2 - \lambda + 2 - 2\lambda = 0$ $7 + 3\lambda = 0 \quad \Rightarrow \quad \lambda = -\frac{7}{3}$.
4. Check Π_2: its normal is $(2, 1, -1)$. Dot with $\mathbf{n}_3$: $6 - 1 - 2 = 3 \neq 0$, so Π_2 is not perpendicular

to Π_3.

5. Substitute $\lambda = -\frac{7}{3}$: $(x-2y+z-3)-\frac{7}{3}(2x+y-z-1) = 0$ Multiply by 3: $3(x-2y+z-3)-7(2x+y-z-1) = 0$ $3x - 6y + 3z - 9 - 14x - 7y + 7z + 7 = 0$ $-11x - 13y + 10z - 2 = 0$ or $11x + 13y - 10z + 2 = 0$.

Answer: $\boxed{11x + 13y - 10z = -2}$.

WORKED EXAMPLE 19 **Plane Through the Intersection Line, Parallel to a Given Line**

Find the plane through the intersection of $\Pi_1 : x + y + z = 6$ and $\Pi_2 : 2x - y + 3z = 5$ that is parallel to the line $L' : \frac{x-1}{2} = \frac{y+2}{-1} = \frac{z}{3}$.

SOLUTION

Solution.

1. $\Pi_1 : x + y + z - 6 = 0, \quad \Pi_2 : 2x - y + 3z - 5 = 0.$
2. Family: $(x + y + z - 6) + \lambda(2x - y + 3z - 5) = 0$. Normal: $\mathbf{n}(\lambda) = (1 + 2\lambda, 1 - \lambda, 1 + 3\lambda)$.
3. Direction of L': $\mathbf{v} = (2, -1, 3)$. Parallel $\Rightarrow \mathbf{n}(\lambda) \cdot \mathbf{v} = 0$: $(1 + 2\lambda)(2) + (1 - \lambda)(-1) + (1 + 3\lambda)(3) = 0$ $2 + 4\lambda - 1 + \lambda + 3 + 9\lambda = 0$ $4 + 14\lambda = 0 \quad \Rightarrow \quad \lambda = -\frac{2}{7}$.
4. Check Π_2: normal $(2, -1, 3)$ dot $\mathbf{v} = 4 + 1 + 9 = 14 \neq 0$, so Π_2 not parallel.
5. Substitute $\lambda = -\frac{2}{7}$: $(x+y+z-6)-\frac{2}{7}(2x-y+3z-5) = 0$ Multiply by 7: $7(x+y+z-6)-2(2x-y+3z-5) = 0$ $7x + 7y + 7z - 42 - 4x + 2y - 6z + 10 = 0$ $3x + 9y + z - 32 = 0$.

Answer: $\boxed{3x + 9y + z = 32}$.

6.3 Lesson 51 — Planes Through the Point of Intersection of Two Planes

The syllabus mentions "Equation of planes through the point of intersection of two planes". Two planes intersect in a line, not a point. However, if a third plane is introduced, the three planes may intersect in a unique point. In that context, the family method can be adapted: the equation of any plane through the intersection point of three planes $\Pi_1 = 0, \Pi_2 = 0, \Pi_3 = 0$ (provided they meet at a single point) is a linear combination $\alpha\Pi_1 + \beta\Pi_2 + \gamma\Pi_3 = 0$. But the standard GCE technique for "planes through the line of intersection of two planes and a given point" is exactly what we have just done (Lessons 50 and 51 are essentially the same skill).

EXEMPLE **Plane Through the Line of Intersection and a Given Point (Lesson 50/51)**

Find the plane through the line of intersection of $x + 2y - z = 4$ and $2x - y + 3z = 1$ that also passes through $A(3, 1, -1)$.

Family: $(x+2y-z-4)+\lambda(2x-y+3z-1)=0$. Substitute A: $(3+2-(-1)-4)+\lambda(6-1-3-1)=0 \Rightarrow 2+\lambda(1)=0 \Rightarrow \lambda=-2$. Check Π_2: $2(3)-1+3(-1)=6-1-3=2\neq 1$, so Π_2 not valid. Equation: $(x+2y-z-4)-2(2x-y+3z-1)=0 \Rightarrow -3x+4y-7z-2=0$ or $3x-4y+7z=-2$.

6.4 Full GCE-Style Synthesis

A typical Upper Sixth question combines several skills: find the line of intersection of two planes, then find a plane through that line satisfying an extra condition. Let us work through a complete example.

WORKED EXAMPLE 20 **Synthesis: Line of Intersection + Family of Planes**

The planes Π_1 and Π_2 have equations

$$\Pi_1:\ 2x-y+z=4, \qquad \Pi_2:\ x+3y-2z=1.$$

1. Find a vector equation of the line L of intersection of Π_1 and Π_2.
2. Find the Cartesian equation of the plane Π that contains L and is perpendicular to the plane $\Pi_3:\ 4x-2y+z=7$.

SOLUTION

Solution.

1. Line of intersection. Solve the system:

$$\begin{cases} 2x-y+z=4 & (1)\\ x+3y-2z=1 & (2)\end{cases}$$

Let $z=t$ (parameter). From (1): $2x-y=4-t$. From (2): $x+3y=1+2t$. Solve for x,y in terms of t: Multiply (2) by 2: $2x+6y=2+4t$. Subtract (1): $7y=(2+4t)-(4-t)=-2+5t \Rightarrow y=\frac{5t-2}{7}$. Then $x=1+2t-3y=1+2t-\frac{15t-6}{7}=\frac{7+14t-15t+6}{7}=\frac{13-t}{7}$. So $L:\ \mathbf{r}=\begin{pmatrix}13/7\\-2/7\\0\end{pmatrix}+t\begin{pmatrix}-1/7\\5/7\\1\end{pmatrix}$. A neater direction vector is $\mathbf{d}=(-1,5,7)$ (multiply by 7). A point on L is $A\left(\frac{13}{7},-\frac{2}{7},0\right)$. Vector equation: $\mathbf{r}=\begin{pmatrix}13/7\\-2/7\\0\end{pmatrix}+\mu\begin{pmatrix}-1\\5\\7\end{pmatrix}$.

2. Plane through L perpendicular to Π_3. Family: $(2x-y+z-4)+\lambda(x+3y-2z-1)=0$. Normal of family: $\mathbf{n}(\lambda)=(2+\lambda,-1+3\lambda,1-2\lambda)$. Π_3 has normal $\mathbf{n}_3=(4,-2,1)$. Perpendicular $\Rightarrow \mathbf{n}(\lambda)\cdot\mathbf{n}_3=0$: $(2+\lambda)4+(-1+3\lambda)(-2)+(1-2\lambda)(1)=0\ 8+4\lambda+2-6\lambda+1-2\lambda=0\ 11-4\lambda=0 \quad\Rightarrow\quad \lambda=\frac{11}{4}$. Check

Π_2: normal $(1,3,-2)$ dot $\mathbf{n}_3 = 4-6-2 = -4 \neq 0$, so Π_2 not perpendicular. Substitute $\lambda = \frac{11}{4}$: $(2x - y + z - 4) + \frac{11}{4}(x + 3y - 2z - 1) = 0$ Multiply by 4: $4(2x - y + z - 4) + 11(x + 3y - 2z - 1) = 0$ $8x - 4y + 4z - 16 + 11x + 33y - 22z - 11 = 0$ $19x + 29y - 18z - 27 = 0$.

Answers:

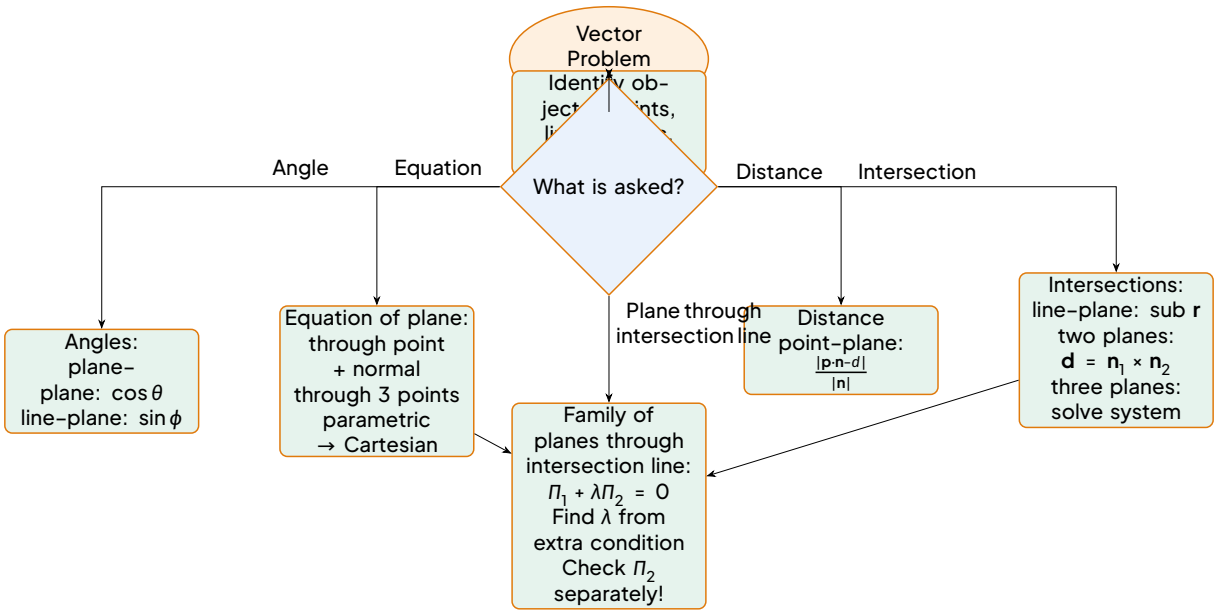
1. L : $\mathbf{r} = \begin{pmatrix} 13/7 \\ -2/7 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} -1 \\ 5 \\ 7 \end{pmatrix}$ (or equivalent).

2. Π : $\boxed{19x + 29y - 18z = 27}$.

6.5 Revision Summary — All Formulas of the Chapter

The table below collects every key formula and result from the whole Vectors chapter (Sections 1–6). Keep it handy for revision.

Topic	Key Formulas / Results
1. Equations of a Plane	Vector: $\mathbf{r} \cdot \mathbf{n} = d$ or $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b} + \mu\mathbf{c}$ Cartesian: $ax + by + cz = d$ where $\mathbf{n} = (a, b, c)$ Through point $\mathbf{a}$ with normal $\mathbf{n}$: $(\mathbf{r} - \mathbf{a}) \cdot \mathbf{n} = 0$ Through three non-collinear points A, B, C: $(\mathbf{r}-\mathbf{a})\cdot((\mathbf{b}-\mathbf{a})\times(\mathbf{c}-\mathbf{a})) = 0$
2. Parametric Form	$\mathbf{r} = \mathbf{a} + \lambda\mathbf{b} + \mu\mathbf{c}$ ($\mathbf{b}, \mathbf{c}$ parallel to plane, not parallel to each other) Cartesian from parametric: eliminate λ, μ or use $\mathbf{n} = \mathbf{b} \times \mathbf{c}$
3. Perpendicular Distance	Point $P(\mathbf{p})$ to plane $\mathbf{r} \cdot \mathbf{n} = d$: $\dfrac{\lvert\mathbf{p} \cdot \mathbf{n} - d\rvert}{\lvert\mathbf{n}\rvert}$ Cartesian: $\dfrac{\lvert ax_0 + by_0 + cz_0 - d\rvert}{\sqrt{a^2 + b^2 + c^2}}$
4. Angles	Angle between planes Π_1, Π_2: $\cos\theta = \dfrac{\lvert\mathbf{n}_1 \cdot \mathbf{n}_2\rvert}{\lvert\mathbf{n}_1\rvert\lvert\mathbf{n}_2\rvert}$ Angle between line $\mathbf{r} = \mathbf{a} + t\mathbf{v}$ and plane $\mathbf{r} \cdot \mathbf{n} = d$: $\sin\phi = \dfrac{\lvert\mathbf{v} \cdot \mathbf{n}\rvert}{\lvert\mathbf{v}\rvert\lvert\mathbf{n}\rvert}$
5. Intersections	Line $\mathbf{r} = \mathbf{a}+t\mathbf{v}$ and plane $\mathbf{r}\cdot\mathbf{n} = d$: substitute, solve for t, find point. Two planes $\mathbf{r} \cdot \mathbf{n}_1 = d_1$, $\mathbf{r} \cdot \mathbf{n}_2 = d_2$: direction $\mathbf{d} = \mathbf{n}_1 \times \mathbf{n}_2$; find one common point. Three planes: solve 3 × 3 system; unique point / line / no intersection (prism).
6. Family of Planes	Through intersection of $\Pi_1 = 0$ and $\Pi_2 = 0$: $\Pi_1 + \lambda\Pi_2 = 0$ ($\lambda \in \mathbb{R}$). **Extra conditions:** – Through point P: substitute P to find λ. – Perpendicular to Π_3: $(\mathbf{n}_1 + \lambda\mathbf{n}_2) \cdot \mathbf{n}_3 = 0$. – Parallel to line $\mathbf{v}$: $(\mathbf{n}_1 + \lambda\mathbf{n}_2) \cdot \mathbf{v} = 0$. **Always check Π_2 separately!**



Flowchart for solving GCE Vector problems — follow the arrows from the question type to the appropriate tool.

6.6 Common Mistakes and Exam Tips

WARNING **Forgetting to Check Π_2**

The family $\Pi_1 + \lambda\Pi_2 = 0$ never gives Π_2 itself. If the extra condition is satisfied by Π_2 (e.g. the given point lies on Π_2, or Π_2 is perpendicular to Π_3), then Π_2 is a valid answer and you will lose marks if you only give the λ-plane.

WARNING **Arithmetic Errors in λ**

Substituting the point or computing the dot product for perpendicularity/parallelism involves signed numbers. Work carefully, keep fractions exact, and double-check your algebra before simplifying the final equation.

WARNING **Not Simplifying to Integer Coefficients**

GCE marking schemes usually expect the final Cartesian equation with integer coefficients (and preferably the constant term positive). After finding λ, multiply through by the denominator and simplify.

KEY POINTS **Exam Tip — The Family Method is a "Gift" Question**

In many GCE papers, the last 4–6 marks of a vector question are awarded for correctly using the family of planes. The steps are routine:

1. Write $\Pi_1 = 0, \Pi_2 = 0$.
2. Form $\Pi_1 + \lambda\Pi_2 = 0$.
3. Use the condition to find λ.
4. Check Π_2.
5. Substitute and simplify.

Practise this pattern until it becomes automatic.

DID YOU KNOW? **Historical Note**

The idea of a "pencil of planes" (family through a common line) was studied by projective geometers like Poncelet and Möbius in the 19th century. It is a beautiful example of how a linear combination of equations preserves the common solutions — a principle that underlies much of linear algebra and algebraic geometry.

EXERCISE 6 **Consolidation Exercises**

1. Find the equation of the plane through the intersection of $x - 2y + 3z = 4$ and $2x + y - z = 5$ that passes through $(1, 1, 1)$.
2. Determine the plane through the line of intersection of $3x - y + 2z = 1$ and $x + 4y - z = 3$ which is perpendicular to $2x - y + 3z = 6$.
3. The planes $\Pi_1 : x + y + z = 3$ and $\Pi_2 : 2x - y + 3z = 4$ intersect in line L. Find the plane containing L that is parallel to the line $\frac{x}{1} = \frac{y}{-2} = \frac{z}{2}$.
4. (Synthesis) Given $\Pi_1 : 2x - y + z = 5$, $\Pi_2 : x + 2y - 3z = 1$, $\Pi_3 : 3x - y + 2z = 4$.
   (a) Find the line of intersection of Π_1 and Π_2.
   (b) Find the plane through this line that is perpendicular to Π_3.
5. A plane Π passes through the intersection of $x + 2y - z = 2$ and $3x - y + 2z = 5$. If Π also passes through the point $(0, 0, 0)$, find its equation. (Don't forget to check Π_2!)

SOLUTION **Exercise 1**

Family: $(x - 2y + 3z - 4) + \lambda(2x + y - z - 5) = 0$. Substitute $(1,1,1)$: $(1 - 2 + 3 - 4) + \lambda(2 + 1 - 1 - 5) = -2 + \lambda(-3) = 0 \Rightarrow \lambda = -\frac{2}{3}$. Check Π_2: $2 + 1 - 1 = 2 \neq 5$, so Π_2 not valid. Equation: $(x-2y+3z-4)-\frac{2}{3}(2x+y-z-5) = 0 \Rightarrow 3x-6y+9z-12-4x-2y+2z+10 = 0 \Rightarrow -x-8y+11z-2 = 0$ or $x + 8y - 11z = -2$.

SOLUTION **Exercise 2**

Family: $(3x - y + 2z - 1) + \lambda(x + 4y - z - 3) = 0$. Normal: $(3 + \lambda, -1 + 4\lambda, 2 - \lambda)$. Perpendicular to $2x - y + 3z = 6 \Rightarrow (3 + \lambda)2 + (-1 + 4\lambda)(-1) + (2 - \lambda)3 = 0\ 6 + 2\lambda + 1 - 4\lambda + 6 - 3\lambda = 0 \Rightarrow 13 - 5\lambda = 0 \Rightarrow \lambda = 13/5$. Check Π_2: normal $(1,4,-1)$ dot $(2,-1,3) = 2 - 4 - 3 = -5 \neq 0$. Equation: $(3x - y + 2z - 1) + \frac{13}{5}(x + 4y - z - 3) = 0 \Rightarrow 5(3x - y + 2z - 1) + 13(x + 4y - z - 3) = 0$ $15x - 5y + 10z - 5 + 13x + 52y - 13z - 39 = 0 \Rightarrow 28x + 47y - 3z - 44 = 0$.

SOLUTION **Exercise 3**

Family: $(x + y + z - 3) + \lambda(2x - y + 3z - 4) = 0$. Normal: $(1 + 2\lambda, 1 - \lambda, 1 + 3\lambda)$. Parallel to line with direction $(1,-2,2) \Rightarrow (1 + 2\lambda)(1) + (1 - \lambda)(-2) + (1 + 3\lambda)(2) = 0\ 1 + 2\lambda - 2 + 2\lambda + 2 + 6\lambda = 0 \Rightarrow 1 + 10\lambda = 0 \Rightarrow \lambda = -0.1$. Check Π_2: normal $(2,-1,3)$ dot $(1,-2,2) = 2 + 2 + 6 = 10 \neq 0$. Equation: $(x + y + z - 3) - 0.1(2x - y + 3z - 4) = 0 \Rightarrow 10(x + y + z - 3) - (2x - y + 3z - 4) = 0$ $10x + 10y + 10z - 30 - 2x + y - 3z + 4 = 0 \Rightarrow 8x + 11y + 7z - 26 = 0$.

SOLUTION **Exercise 4**

(a) Solve $2x - y + z = 5$, $x + 2y - 3z = 1$. Let $z = t$: $2x - y = 5 - t$, $x + 2y = 1 + 3t$. $x = 2 + ?$ Solve: multiply first by 2: $4x - 2y = 10 - 2t$. Add second: $5x = 11 + t \Rightarrow x = (11 + t)/5$. $y = (1 + 3t - x)/2 = (1 + 3t - (11 + t)/5)/2 = (5 + 15t - 11 - t)/10 = (14t - 6)/10 = (7t - 3)/5$. Line: $\mathbf{r} = \begin{pmatrix} 11/5 \\ -3/5 \\ 0 \end{pmatrix} + t\begin{pmatrix} 1/5 \\ 7/5 \\ 1 \end{pmatrix}$ or direction $(1,7,5)$. (b) Family: $(2x-y+z-5)+\lambda(x+2y-3z-1) = 0$. Normal: $(2+\lambda, -1+2\lambda, 1-3\lambda)$. Perpendicular to Π_3 (normal $(3,-1,2)$): $(2+\lambda)3+(-1+2\lambda)(-1)+(1-3\lambda)2 = 0$ $6 + 3\lambda + 1 - 2\lambda + 2 - 6\lambda = 0 \Rightarrow 9 - 5\lambda = 0 \Rightarrow \lambda = 9/5$. Check Π_2: normal $(1,2,-3)$ dot $(3,-1,2) = 3-2-6 = -5 \neq 0$. Equation: $(2x-y+z-5)+\frac{9}{5}(x+2y-3z-1) = 0 \Rightarrow 5(2x-y+z-5)+9(x+2y-3z-1) = 0$ $10x - 5y + 5z - 25 + 9x + 18y - 27z - 9 = 0 \Rightarrow 19x + 13y - 22z - 34 = 0$.

SOLUTION **Exercise 5**

Family: $(x+2y-z-2)+\lambda(3x-y+2z-5)=0$. Passes through $(0,0,0)$: $-2+\lambda(-5)=0 \Rightarrow \lambda=-2/5$. Check Π_2: $3(0)-0+2(0)=0\neq 5$, so Π_2 not valid. Equation: $(x+2y-z-2)-\frac{2}{5}(3x-y+2z-5)=0 \Rightarrow 5x+10y-5z-10-6x+2y-4z+10=0 \Rightarrow -x+12y-9z=0$ or $x-12y+9z=0$.

This concludes the Vectors chapter. You now have a complete toolkit: equations of planes in all forms, distances, angles, intersections, and the family of planes technique. Regular practice with past GCE papers will cement these skills. Good luck in your preparation!

7 INTEGRATION ACTIVITY

8 VECTORS: INTEGRATION ACTIVITY

8.1 Aims of the activity

This integration activity closes the chapter on vectors and planes by bringing together, in one realistic engineering situation, every major skill you have developed:

- finding the **Cartesian equation of a plane** through three given points;
- checking whether a point lies on a plane;
- computing the **angle between two planes** using their normal vectors;
- finding the **point of intersection of a line and a plane**;
- computing the **perpendicular distance from a point to a plane** and interpreting it in a practical decision;
- determining the **line of intersection of two planes** and writing its vector equation.

Beyond the calculations, the activity trains you to model a real object (a workshop roof) with mathematical objects (planes and lines), to organise group work, to present a solution clearly on a flipchart, and to check each step against the formulas of the chapter — exactly the kind of structured reasoning expected at the GCE Advanced Level.

8.2 Situation

A Government Technical College in Buea, on the slopes of Mount Cameroon, is constructing a rectangular workshop for its carpentry section. Because Buea receives heavy rainfall for much of the year, the roof must slope so that rainwater runs off quickly into a gutter.

The architect sets up a three-dimensional coordinate system, with the unit being the metre. The concrete floor of the workshop is the horizontal plane $z = 0$. The four top corners of the walls are

$$A(0,0,3), \qquad B(8,0,3), \qquad C(8,6,4), \qquad D(0,6,4).$$

Notice that the front wall (through A and B) is 3 m high, while the back wall (through C and D) is 4 m high: the roof therefore rises from front to back, and the gutter will be fixed along the front edge AB.

The sloping **roof panel** is the plane Π_1 passing through the three points A, B and C.

During construction, the site engineer faces five questions:

1. What is the Cartesian equation of the roof plane Π_1? The corner D was designed to lie on the same panel — does it really?
2. What angle does the roof make with the horizontal floor? (The carpenter needs this pitch angle to cut the wooden trusses.)

3. A vertical steel support cable is fixed to the floor at the point $(4, 3, 0)$ and rises straight up. It is modelled by the line
$$\vec{r} = \begin{pmatrix} 4 \\ 3 \\ 0 \end{pmatrix} + t \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \qquad t \in \mathbb{R}.$$
At what point does the cable meet the underside of the roof, and what length of cable must be ordered?

4. A lamp hangs from a hook at the point $P(4, 3, 5)$. Safety rules require a clearance of at least 1.5 m between the lamp and the roof surface. Is the rule respected?

5. A second roof panel, on the other side of the building, has equation $2x - 3z + 6 = 0$. The **gutter** is installed along the line where the two panels meet. What is the vector equation of this gutter line?

The class works in groups of four. Each group writes its full solution on a flipchart, then presents it; the other groups check every step against the formulas of the chapter (normal vector, scalar product, distance formula, line–plane intersection).

8.3 Tasks

1. **Equation of the roof.** Compute the vectors $\vec{AB}$ and $\vec{AC}$. Deduce a normal vector $\vec{n} = \vec{AB} \times \vec{AC}$ to the plane Π_1. Hence find the Cartesian equation of Π_1 in the form $ax + by + cz + d = 0$. Verify that the coordinates of D satisfy this equation.

2. **Pitch of the roof.** Write down a normal vector to the floor plane $z = 0$. Using the scalar product formula for the angle between two planes, find the acute angle between the roof and the floor, giving your answer to the nearest 0.1°.

3. **The support cable.** Substitute the parametric coordinates $(4, 3, t)$ of a point of the cable into the equation of Π_1 to find the value of t at the meeting point M. State the coordinates of M and deduce the length of cable needed from the floor to the roof.

4. **The lamp clearance.** Use the perpendicular distance formula
$$d = \frac{|ax_0 + by_0 + cz_0 + d|}{\sqrt{a^2 + b^2 + c^2}}$$
to find the distance from $P(4, 3, 5)$ to Π_1, correct to 2 decimal places. Decide, with justification, whether the 1.5 m clearance rule is respected.

5. **The gutter line.** The gutter lies on both Π_1 and the second panel $\Pi_2 : 2x - 3z + 6 = 0$. Solve the two equations simultaneously (introduce a parameter λ) to obtain the vector equation of the line of intersection. Give a direction vector of the gutter and the coordinates of one point on it.

6. **Synthesis.** On your flipchart, draw a labelled sketch of the workshop showing the floor, the roof plane, the cable, the lamp and the gutter, and summarise all five answers in a results table.

8.4 Model Answer

1. Cartesian equation of the roof plane.

From the coordinates of A, B and C:

$$\vec{AB} = \begin{pmatrix} 8 \\ 0 \\ 0 \end{pmatrix}, \qquad \vec{AC} = \begin{pmatrix} 8 \\ 6 \\ 1 \end{pmatrix}.$$

A normal vector to Π_1 is

$$\vec{n} = \vec{AB} \times \vec{AC} = \begin{pmatrix} 8 \\ 0 \\ 0 \end{pmatrix} \times \begin{pmatrix} 8 \\ 6 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \cdot 1 - 0 \cdot 6 \\ 0 \cdot 8 - 8 \cdot 1 \\ 8 \cdot 6 - 0 \cdot 8 \end{pmatrix} = \begin{pmatrix} 0 \\ -8 \\ 48 \end{pmatrix},$$

which we simplify to $\vec{n} = (0, 1, -6)$ after dividing by -8. The plane therefore has equation

$$y - 6z + d = 0.$$

Substituting $A(0, 0, 3)$: $0 - 18 + d = 0$, so $d = 18$. Hence

$$\boxed{\Pi_1 : \; y - 6z + 18 = 0.}$$

Check for $D(0, 6, 4)$: $6 - 24 + 18 = 0$. ✓ So D lies on the roof plane, as the architect intended.

2. Pitch of the roof.

The floor $z = 0$ has normal vector $\vec{n}_0 = (0, 0, 1)$. The angle θ between the planes equals the acute angle between their normals:

$$\cos\theta = \frac{|\vec{n} \cdot \vec{n}_0|}{|\vec{n}|\,|\vec{n}_0|} = \frac{|-6|}{\sqrt{0^2 + 1^2 + (-6)^2}\,\sqrt{1}} = \frac{6}{\sqrt{37}}.$$

Therefore

$$\theta = \cos^{-1}\!\left(\frac{6}{\sqrt{37}}\right) \approx \cos^{-1}(0.9864) \approx 9.6^\circ.$$

The roof rises at about 9.6° to the horizontal — a gentle slope, sufficient for rainwater run-off.

> **WARNING** **Angle between planes**
>
> Always take the acute angle (or right angle) between planes. The formula uses the absolute value of the scalar product of the normals. If the computed angle is obtuse ($> 90^\circ$), subtract it from 180°.

3. The support cable.

A point of the cable has coordinates $(4, 3, t)$. Substituting into $y - 6z + 18 = 0$:

$$3 - 6t + 18 = 0 \quad \Longrightarrow \quad 6t = 21 \quad \Longrightarrow \quad t = \tfrac{7}{2} = 3.5.$$

The cable meets the roof at

$$M(4,\ 3,\ 3.5).$$

Since the cable starts at the floor point $(4, 3, 0)$ and rises vertically, the length needed is simply the height of M:

$$\text{length} = 3.5\ \text{m}.$$

The engineer should order a cable of at least 3.5 m (plus a small allowance for fastening).

4. The lamp clearance.

The distance from $P(4, 3, 5)$ to $\Pi_1 : y - 6z + 18 = 0$ is

$$d = \frac{|3 - 6(5) + 18|}{\sqrt{0^2 + 1^2 + (-6)^2}} = \frac{|3 - 30 + 18|}{\sqrt{37}} = \frac{9}{\sqrt{37}} \approx \frac{9}{6.0828} \approx 1.48\ \text{m}.$$

Since 1.48 m < 1.5 m, the clearance rule is **not respected**: the lamp is about 2 cm too close to the roof surface. The hook must be lowered slightly (or the lamp repositioned) before the workshop is approved.

WARNING **Interpreting the distance**

The perpendicular distance formula gives the shortest distance from the point to the plane. Here the vertical height of P above the floor is 5 m, but the roof above P is at height $z = \frac{3+18}{6} = 3.5$ m, so P is actually 1.5 m vertically above the roof — yet only 1.48 m away perpendicularly. Always use the perpendicular distance for clearance checks, and note that P lies on the opposite side of the plane from the origin (the sign of $y-6z+18$ at P is negative).

5. Vector equation of the gutter line.

The gutter lies on both planes:

$$\begin{cases} y - 6z + 18 = 0 & (\Pi_1) \\ 2x - 3z + 6 = 0 & (\Pi_2) \end{cases}$$

Let $z = \lambda$, where $\lambda \in \mathbb{R}$. Then from Π_1: $y = 6\lambda - 18$, and from Π_2: $2x = 3\lambda - 6$, so $x = \frac{3}{2}\lambda - 3$. Hence

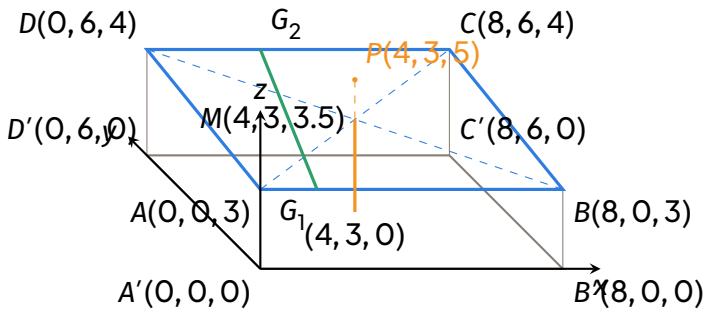
$$\vec{r} = \begin{pmatrix} -3 \\ -18 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} \frac{3}{2} \\ 6 \\ 1 \end{pmatrix}, \qquad \text{or, clearing fractions with } \lambda = 2\mu, \quad \vec{r} = \begin{pmatrix} -3 \\ -18 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 3 \\ 12 \\ 2 \end{pmatrix},\ \mu \in \mathbb{R}.$$

Check of the direction vector: it must be perpendicular to both normals:

$$(3, 12, 2) \cdot (0, 1, -6) = 12 - 12 = 0, \qquad (3, 12, 2) \cdot (2, 0, -3) = 6 - 6 = 0. \checkmark$$

So the gutter runs along the line through $(-3, -18, 0)$ with direction $(3, 12, 2)$; within the actual building ($0 \le x \le 8$, $0 \le y \le 6$), it is the segment where the two panels physically meet, joining the points $(1.5, 0, 3)$ on edge AB and $(3, 6, 4)$ on edge CD.

6. Summary and sketch.



Isometric sketch of the workshop: floor (grey), roof (blue), vertical cable (orange), lamp P, gutter (green). The gutter segment G_1G_2 lies on the intersection of the two roof panels.

Quantity	Result
Roof plane Π_1	$y - 6z + 18 = 0$; D lies on it
Pitch angle roof/floor	$\theta \approx 9.6^\circ$
Cable meets roof at	$M(4,\ 3,\ 3.5)$; cable length = 3.5 m
Lamp clearance	$d = \dfrac{9}{\sqrt{37}} \approx 1.48$ m < 1.5 m: rule not respected
Gutter line	$\vec{r} = (-3, -18, 0) + \mu(3, 12, 2)$, $\mu \in \mathbb{R}$; segment $G_1(1.5, 0, 3)$ to $G_2(3, 6, 4)$

KEY POINTS **What this activity consolidates**

One authentic situation has mobilised the whole chapter: the cross product gives a normal vector and hence the Cartesian equation of a plane; the scalar product of normals gives the angle between planes; substitution of a parametric line into the plane equation gives the line–plane intersection; the formula $\frac{|ax_0+by_0+cz_0+d|}{\sqrt{a^2+b^2+c^2}}$ gives the point–plane distance; and solving two plane equations simultaneously gives the line of intersection of two planes. In the examination, present each step exactly as here: formula first, substitution second, conclusion with units and interpretation last.

9 METHODS AND WORKED EXAMPLES

9.1 Method 1: Finding the Cartesian Equation of a Plane

A plane in space is completely determined as soon as we know one of its points and the direction perpendicular to it. This perpendicular direction is carried by a **normal vector**. Observe first a concrete fact: if $\vec{n}$ is normal to the plane Π and A is a fixed point of Π, then a point P lies on Π if and only if $\vec{AP} \perp \vec{n}$, that is, $\vec{AP} \cdot \vec{n} = 0$. Writing $\vec{r} = \vec{OP}$ and $\vec{a} = \vec{OA}$, this becomes $(\vec{r} - \vec{a}) \cdot \vec{n} = 0$, i.e.

$$\vec{r} \cdot \vec{n} = \vec{a} \cdot \vec{n}.$$

This is the **vector equation of the plane**. Expanding the scalar products with $\vec{n} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$ and $\vec{r} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ gives the **Cartesian equation** $ax + by + cz = d$ where $d = \vec{a} \cdot \vec{n}$.

DEFINITION **Normal vector; equations of a plane**

A **normal vector** to a plane Π is any non-zero vector perpendicular to every line lying in Π.

If $A(x_0, y_0, z_0) \in \Pi$ and $\vec{n} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$ is a normal vector, then:

- vector (scalar product) form: $\vec{r} \cdot \vec{n} = \vec{a} \cdot \vec{n}$;
- Cartesian form: $ax + by + cz = d$ with $d = ax_0 + by_0 + cz_0$.

PROPERTY **Reading information from the Cartesian equation**

For the plane $\Pi : ax + by + cz = d$:

- $\vec{n} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$ is a normal vector to Π;
- two planes are parallel if and only if their normal vectors are parallel (one is a scalar multiple of the other); they are coincident if, in addition, the equations are proportional;
- two planes are perpendicular if and only if $\vec{n}_1 \cdot \vec{n}_2 = 0$.

(Proof not examinable; these follow directly from the definition of the scalar product.)

METHOD **Cartesian equation of a plane from a point and a normal vector**

To find the Cartesian equation of a plane Π:

1. Identify a **normal vector** $\vec{n} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$ to the plane (given, or obtained as a vector product of two directions parallel to the plane).
2. Write the provisional equation $ax + by + cz = d$.
3. Substitute the coordinates of a known point $A(x_0, y_0, z_0)$ of the plane to find $d = ax_0 + by_0 + cz_0$.
4. Conclude: $\Pi : ax + by + cz = d$. Equivalently, the vector form is $\vec{r} \cdot \vec{n} = \vec{a} \cdot \vec{n}$.

Reflex: the coefficients of x, y, z are always the components of a normal vector. Two planes are parallel if and only if their normal vectors are parallel.

WORKED EXAMPLE 21 **Plane through a point with a given normal**

Find the Cartesian equation of the plane Π which passes through the point $A(2, -1, 3)$ and is perpendicular to the vector $\vec{n} = \begin{pmatrix} 1 \\ 4 \\ -2 \end{pmatrix}$.

SOLUTION

Step 1. The normal vector is $\vec{n} = \begin{pmatrix} 1 \\ 4 \\ -2 \end{pmatrix}$, so the equation has the form

$$x + 4y - 2z = d.$$

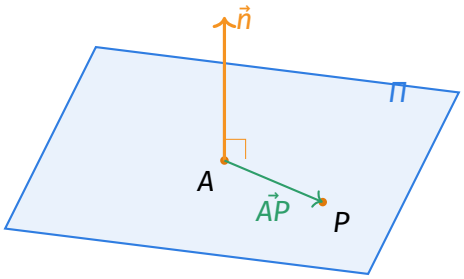
Step 2. Since $A(2, -1, 3)$ lies on Π:

$$d = (2) + 4(-1) - 2(3) = 2 - 4 - 6 = -8.$$

Step 3. Hence

$$\Pi : x + 4y - 2z = -8.$$

Check: substituting A: $2 + 4(-1) - 2(3) = -8$. ✓ The equation is satisfied.



A plane Π through A with normal vector $\vec{n}$: $P \in \Pi \iff \vec{AP} \cdot \vec{n} = 0$.

9.2 Method 2: Plane Through Three Non-Collinear Points

Three points always lie in some plane; but they determine a **unique** plane only when they are not on the same straight line. This is why the collinearity test must come first.

METHOD **Cartesian equation of the plane through three points**

Given three non-collinear points A, B, C:

1. First check the points are not collinear: verify that $\vec{AB}$ and $\vec{AC}$ are not parallel (no scalar k with $\vec{AB} = k\vec{AC}$). If they are parallel, the three points lie on a line and do not define a unique plane.
2. Form the two direction vectors $\vec{AB}$ and $\vec{AC}$.
3. Compute the normal vector $\vec{n} = \vec{AB} \times \vec{AC}$ (vector product): it is perpendicular to both, hence to the plane.
4. Write $ax + by + cz = d$ and find d using any one of the three points.
5. Verify that all three points satisfy the final equation.

Reflex: the vector product $\begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} \times \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} = \begin{pmatrix} a_2b_3 - a_3b_2 \\ a_3b_1 - a_1b_3 \\ a_1b_2 - a_2b_1 \end{pmatrix}$.

WORKED EXAMPLE 22 **Plane through three points**

Find the Cartesian equation of the plane containing the points $A(1, 0, 2)$, $B(2, 1, 1)$ and $C(-1, 2, 1)$.

SOLUTION

Step 1. Direction vectors.

$$\vec{AB} = \begin{pmatrix} 2-1 \\ 1-0 \\ 1-2 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}, \qquad \vec{AC} = \begin{pmatrix} -1-1 \\ 2-0 \\ 1-2 \end{pmatrix} = \begin{pmatrix} -2 \\ 2 \\ -1 \end{pmatrix}.$$

These are not parallel (no k satisfies $1 = -2k$ and $1 = 2k$ simultaneously), so A, B, C are not collinear and define a unique plane.
Step 2. Normal vector.

$$\vec{n} = \vec{AB} \times \vec{AC} = \begin{pmatrix} (1)(-1) - (-1)(2) \\ (-1)(-2) - (1)(-1) \\ (1)(2) - (1)(-2) \end{pmatrix} = \begin{pmatrix} -1+2 \\ 2+1 \\ 2+2 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \\ 4 \end{pmatrix}.$$

Step 3. Find d. The equation is $x + 3y + 4z = d$. Using $A(1, 0, 2)$:

$$d = 1 + 3(0) + 4(2) = 9.$$

Step 4. Conclusion and verification.

$$\Pi : \; x + 3y + 4z = 9.$$

Check $B(2, 1, 1)$: $2 + 3 + 4 = 9$ ✓. Check $C(-1, 2, 1)$: $-1 + 6 + 4 = 9$ ✓.

WARNING **Three collinear points**

If the three given points are collinear, $\vec{AB} \times \vec{AC} = \vec{0}$ and the method collapses: infinitely many planes contain a given line. Always test collinearity first. At the GCE, a question may deliberately give collinear points and ask you to comment.

WORKED EXAMPLE 23 **The collinear case**

Show that the points $A(1, 2, 0)$, $B(3, 4, 2)$ and $C(5, 6, 4)$ do not determine a unique plane.

SOLUTION

$\vec{AB} = \begin{pmatrix} 2 \\ 2 \\ 2 \end{pmatrix}$ and $\vec{AC} = \begin{pmatrix} 4 \\ 4 \\ 4 \end{pmatrix} = 2\vec{AB}$. The vectors are parallel, so A, B, C are collinear: they all lie on

the line through A with direction $\begin{pmatrix}1\\1\\1\end{pmatrix}$. Infinitely many planes contain this line, so no unique plane is determined. (Note that $\vec{AB} \times \vec{AC} = \vec{0}$, confirming the failure of the cross-product method.)

9.3 Method 3: Parametric (Vector) Form of the Equation of a Plane

Just as a line is described by one point and one direction, a plane is described by one point and **two** independent directions. Moving from A along any combination of $\vec{u}$ and $\vec{v}$ keeps us inside the plane: this is the geometric content of the parametric form.

DEFINITION **Parametric vector equation of a plane**

The plane through A parallel to the non-parallel vectors $\vec{u}$ and $\vec{v}$ has equation

$$\vec{r} = \vec{a} + \lambda\vec{u} + \mu\vec{v}, \qquad \lambda, \mu \in \mathbb{R}.$$

Each pair (λ, μ) gives exactly one point of the plane.

METHOD **Parametric form of a plane; converting between forms**

A plane through A with direction vectors $\vec{u}$ and $\vec{v}$ (non-parallel) has **parametric vector equation**

$$\vec{r} = \vec{a} + \lambda\vec{u} + \mu\vec{v}, \qquad \lambda, \mu \in \mathbb{R}.$$

1. To obtain the parametric form from three points A, B, C: take $\vec{u} = \vec{AB}$, $\vec{v} = \vec{AC}$.
2. To pass from parametric to Cartesian form: compute $\vec{n} = \vec{u} \times \vec{v}$, then $d = \vec{a} \cdot \vec{n}$, giving $\vec{r} \cdot \vec{n} = d$, i.e. $ax + by + cz = d$.
3. To pass from Cartesian to parametric form: find one point of the plane (set two coordinates to convenient values) and two independent vectors perpendicular to $\vec{n}$.

Reflex: a point P lies on the plane if and only if there exist λ, μ giving its position vector; solve the resulting system of three equations in two unknowns and check consistency.

WORKED EXAMPLE 24 **Parametric form and conversion**

The plane Π contains $A(1, 1, 0)$, $B(0, 2, 1)$ and $C(2, 0, 3)$.

(a) Write a vector equation of Π in parametric form.

(b) Deduce the Cartesian equation of Π.

(c) Determine whether the point $P(3, -1, 5)$ lies on Π.

SOLUTION

(a) $\vec{AB} = \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}$, $\vec{AC} = \begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix}$. Hence

$$\vec{r} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix}, \qquad \lambda, \mu \in \mathbb{R}.$$

(b) Normal vector:

$$\vec{n} = \vec{AB} \times \vec{AC} = \begin{pmatrix} (1)(3) - (1)(-1) \\ (1)(1) - (-1)(3) \\ (-1)(-1) - (1)(1) \end{pmatrix} = \begin{pmatrix} 4 \\ 4 \\ 0 \end{pmatrix} = 4 \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}.$$

Take $\vec{n} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$. Then $d = 1(1) + 1(1) + 0(0) = 2$, so

$$\Pi : \; x + y = 2.$$

(c) Using the parametric form: we need

$$1 - \lambda + \mu = 3, \qquad 1 + \lambda - \mu = -1, \qquad \lambda + 3\mu = 5.$$

The first equation gives $\mu - \lambda = 2$; the second gives $\lambda - \mu = -2$, i.e. the same condition. From the third, $\lambda = 5 - 3\mu$. Substituting: $\mu - (5 - 3\mu) = 2 \Rightarrow 4\mu = 7 \Rightarrow \mu = \frac{7}{4}, \lambda = 5 - \frac{21}{4} = -\frac{1}{4}$. The system is consistent, so $P \in \Pi$.
Quicker check with (b): $3 + (-1) = 2$ ✓, confirming $P \in \Pi$.

WARNING **Two parameters, not one**

A plane has two parameters (λ and μ); a line has only one. Writing $\vec{r} = \vec{a} + \lambda\vec{u}$ for a plane is a classic error: that is the equation of a line. Also, the two direction vectors must not be parallel, otherwise they span only a line.

9.4 Method 4: Perpendicular Distance from a Point to a Plane

The shortest distance from a point P to a plane Π is measured along the perpendicular from P to Π. If H is the foot of this perpendicular and A is any point of Π, then $\vec{PH}$ is parallel to $\vec{n}$, and projecting $\vec{AP}$ onto the unit normal $\hat{\vec{n}}$ gives $|PH| = |\vec{AP} \cdot \hat{\vec{n}}|$. Expanding this with $\vec{n} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$ and $\Pi : ax + by + cz = d$ yields the formula below.

PROPERTY **Distance from a point to a plane**

The perpendicular distance from $P(x_1, y_1, z_1)$ to the plane $\Pi : ax + by + cz = d$ is

$$\delta = \frac{|ax_1 + by_1 + cz_1 - d|}{\sqrt{a^2 + b^2 + c^2}}.$$

(The derivation sketched above — projection onto the unit normal — is a useful justification but only the formula and its use are required at the GCE.)

METHOD **Distance from a point to a plane**

For the plane $\Pi : ax + by + cz = d$ and the point $P(x_1, y_1, z_1)$:

1. Identify a, b, c, d (write the equation in the form $ax + by + cz - d = 0$ if necessary).
2. Apply the formula
$$\delta = \frac{|ax_1 + by_1 + cz_1 - d|}{\sqrt{a^2 + b^2 + c^2}}.$$
3. If the foot of the perpendicular H is required: parametrise the line through P in direction $\vec{n}$: $\vec{r} = \vec{p} + t\vec{n}$, substitute into the plane equation to find t, then H corresponds to that t.

Reflex: the numerator is the left-hand side of the plane equation evaluated at P, in absolute value; the denominator is the modulus of the normal vector.

WORKED EXAMPLE 25 **Distance and foot of the perpendicular**

Find the perpendicular distance from the point $P(1, 2, 3)$ to the plane $\Pi : 2x - y + 2z = 5$, and determine the coordinates of the foot of the perpendicular from P to Π.

SOLUTION

Step 1. Distance. Here $a = 2, b = -1, c = 2, d = 5$:

$$\delta = \frac{|2(1) - (2) + 2(3) - 5|}{\sqrt{2^2 + (-1)^2 + 2^2}} = \frac{|2 - 2 + 6 - 5|}{\sqrt{9}} = \frac{1}{3}.$$

So the perpendicular distance is $\frac{1}{3}$ units.

Step 2. Foot of the perpendicular. The line through P perpendicular to Π has direction $\vec{n} = \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}$:

$$\vec{r} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + t \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}, \quad \text{i.e. } x = 1 + 2t,\ y = 2 - t,\ z = 3 + 2t.$$

Substitute into $2x - y + 2z = 5$:

$$2(1 + 2t) - (2 - t) + 2(3 + 2t) = 5$$

$$2 + 4t - 2 + t + 6 + 4t = 5 \implies 9t + 6 = 5 \implies t = -\tfrac{1}{9}.$$

Hence

$$H = \left(1 - \tfrac{2}{9},\ 2 + \tfrac{1}{9},\ 3 - \tfrac{2}{9}\right) = \left(\tfrac{7}{9},\ \tfrac{19}{9},\ \tfrac{25}{9}\right).$$

Check: $2\left(\frac{7}{9}\right) - \frac{19}{9} + 2\left(\frac{25}{9}\right) = \frac{14-19+50}{9} = \frac{45}{9} = 5$ ✓. Also $|PH| = |t|\,|\vec{n}| = \frac{1}{9} \times 3 = \frac{1}{3}$, consistent with Step 1.

WARNING **Signs and the absolute value**

- The formula contains $-d$: for $\Pi: \ 2x - y + 2z = 5$ you must compute $2x_1 - y_1 + 2z_1$**-5**, not $+5$.
- Do not forget the absolute value: a distance is never negative.
- If $\delta = 0$, the point lies on the plane.

9.5 Method 5: Angles Between Planes, and Between a Line and a Plane

The angle between two planes is defined as the angle between two lines, one in each plane, both perpendicular to the line of intersection. A moment's thought (or a sketch) shows this equals the angle between the two **normal vectors** — or its supplement; taking the absolute value of the scalar product selects the acute angle. For a line and a plane, the angle is measured between the line and its projection on the plane; this is the **complement** of the angle between the line and the normal, which is why sine replaces cosine.

PROPERTY **Angle formulas**

- Angle between two planes with normals $\vec{n}_1, \vec{n}_2$:

$$\cos\theta = \frac{|\vec{n}_1 \cdot \vec{n}_2|}{|\vec{n}_1|\,|\vec{n}_2|}, \qquad 0 \le \theta \le 90^\circ.$$

- Angle between a line (direction $\vec{d}$) and a plane (normal $\vec{n}$):

$$\sin\varphi = \frac{|\vec{d} \cdot \vec{n}|}{|\vec{d}|\,|\vec{n}|}, \qquad 0 \le \varphi \le 90^\circ.$$

(Proofs not examinable; the formulas must be known and used confidently.)

METHOD **Angles involving planes and lines**

1. Angle between two planes with normals $\vec{n}_1, \vec{n}_2$: this equals the angle between the normals:
$$\cos\theta = \frac{|\vec{n}_1 \cdot \vec{n}_2|}{|\vec{n}_1|\,|\vec{n}_2|}.$$
The modulus ensures the acute angle between the planes.
2. Angle φ between a line (direction $\vec{d}$) and a plane (normal $\vec{n}$): the line makes angle θ with the normal where $\cos\theta = \dfrac{|\vec{d} \cdot \vec{n}|}{|\vec{d}||\vec{n}|}$, and the required angle is the complement:
$$\sin\varphi = \frac{|\vec{d} \cdot \vec{n}|}{|\vec{d}|\,|\vec{n}|}, \qquad \varphi = 90^\circ - \theta.$$

Reflex: plane–plane uses cosine of the normals; line–plane uses sine with the normal. A line is parallel to a plane iff $\vec{d} \cdot \vec{n} = 0$; perpendicular iff $\vec{d}$ is parallel to $\vec{n}$.

WORKED EXAMPLE 26 **Two angle computations**

Consider the planes $\Pi_1 : x + 2y - 2z = 3$ and $\Pi_2 : 2x - y + z = 1$, and the line $L : \vec{r} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} + t\begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}$.

(a) Find the acute angle between Π_1 and Π_2.

(b) Find the angle between L and Π_1.

SOLUTION

(a) The normals are $\vec{n}_1 = \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}$ and $\vec{n}_2 = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$.

$$\vec{n}_1 \cdot \vec{n}_2 = (1)(2) + (2)(-1) + (-2)(1) = 2 - 2 - 2 = -2.$$

$$|\vec{n}_1| = \sqrt{1 + 4 + 4} = 3, \qquad |\vec{n}_2| = \sqrt{4 + 1 + 1} = \sqrt{6}.$$

$$\cos\theta = \frac{|-2|}{3\sqrt{6}} = \frac{2}{3\sqrt{6}} = \frac{2\sqrt{6}}{18} = \frac{\sqrt{6}}{9} \approx 0.2722.$$

$$\theta = \cos^{-1}(0.2722) \approx 74.2^\circ.$$

(b) Direction of L: $\vec{d} = \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}$, $|\vec{d}| = \sqrt{4 + 1 + 4} = 3$.

$$\vec{d} \cdot \vec{n}_1 = (2)(1) + (1)(2) + (2)(-2) = 2 + 2 - 4 = 0.$$

Since $\vec{d} \cdot \vec{n}_1 = 0$, the line is parallel to the plane Π_1: the angle between L and Π_1 is 0°. (Indeed $\sin\varphi = 0$.)

9.6 Method 6: Intersection of a Line and a Plane; Intersection of Two Planes

METHOD **Line–plane and plane–plane intersections**

Line and plane.

1. Write the line in parametric form: $x = x_0 + \lambda a$, $y = y_0 + \lambda b$, $z = z_0 + \lambda c$.
2. Substitute into the plane equation $ax + by + cz = d$ and solve the linear equation in λ.
3. One solution ⇒ one intersection point (substitute back). No solution ⇒ line parallel to plane, not contained in it. Infinitely many ⇒ line lies in the plane.

Two planes.

1. Solve the two Cartesian equations simultaneously: eliminate one variable, set one free variable = λ, express x, y, z in terms of λ.
2. The result is the line of intersection in parametric form; read off a point and a direction vector.
3. Alternative for the direction: $\vec{d} = \vec{n}_1 \times \vec{n}_2$.

4. If $\vec{n}_1 \times \vec{n}_2 = \vec{0}$, the planes are parallel (distinct or coincident).

WORKED EXAMPLE 27 **Line meets plane; two planes meet in a line**

(a) Find the point of intersection of the line $\vec{r} = \begin{pmatrix}2\\1\\0\end{pmatrix} + \lambda\begin{pmatrix}1\\-1\\2\end{pmatrix}$ with the plane Π : $x + 2y - z = 6$.

(b) Find, in vector form, the line of intersection of the planes Π_1 : $x + y + z = 4$ and Π_2 : $2x - y + z = 5$.

SOLUTION

(a) Parametrically: $x = 2 + \lambda$, $y = 1 - \lambda$, $z = 2\lambda$. Substitute into Π:

$$(2 + \lambda) + 2(1 - \lambda) - 2\lambda = 6 \implies 2 + \lambda + 2 - 2\lambda - 2\lambda = 6$$

$$4 - 3\lambda = 6 \implies \lambda = -\tfrac{2}{3}.$$

The intersection point is

$$\left(2 - \tfrac{2}{3}, 1 + \tfrac{2}{3}, -\tfrac{4}{3}\right) = \left(\tfrac{4}{3}, \tfrac{5}{3}, -\tfrac{4}{3}\right).$$

Check: $\frac{4}{3} + 2\left(\frac{5}{3}\right) - \left(-\frac{4}{3}\right) = \frac{4+10+4}{3} = \frac{18}{3} = 6$ ✓.

(b) Subtract the equations: $(2x - y + z) - (x + y + z) = 5 - 4$, i.e.

$$x - 2y = 1 \implies x = 1 + 2y.$$

Let $y = \lambda$. Then $x = 1 + 2\lambda$, and from Π_1: $z = 4 - x - y = 4 - (1 + 2\lambda) - \lambda = 3 - 3\lambda$. Hence

$$\vec{r} = \begin{pmatrix}1\\0\\3\end{pmatrix} + \lambda\begin{pmatrix}2\\1\\-3\end{pmatrix}, \qquad \lambda \in \mathbb{R}.$$

Check of the direction: $\vec{n}_1 \times \vec{n}_2 = \begin{pmatrix}1\\1\\1\end{pmatrix} \times \begin{pmatrix}2\\-1\\1\end{pmatrix} = \begin{pmatrix}(1)(1) - (1)(-1)\\(1)(2) - (1)(1)\\(1)(-1) - (1)(2)\end{pmatrix} = \begin{pmatrix}2\\1\\-3\end{pmatrix}$ ✓. Check the point $(1, 0, 3)$: in Π_1: $1 + 0 + 3 = 4$ ✓; in Π_2: $2 - 0 + 3 = 5$ ✓.

WARNING **The three possible positions of a line and a plane**

After substitution you obtain a linear equation in λ:

- $k\lambda = m$ with $k \neq 0$: unique intersection point;
- $0 \cdot \lambda = m$ with $m \neq 0$ (e.g. $0 = 5$): impossible — the line is parallel to the plane and does not touch it;
- $0 \cdot \lambda = 0$: true for every λ — the line lies entirely in the plane.

Always interpret the algebraic outcome geometrically; the GCE regularly asks for this comment.

9.7 Method 7: Family of Planes Through the Line of Intersection of Two Planes

Why does the combination $\Pi_1 + \lambda\Pi_2 = 0$ work? If a point lies on the line of intersection ℓ, it satisfies both equations, so both brackets vanish and the combination vanishes for every λ: hence every plane of the family contains ℓ. Conversely, the combination is linear in x, y, z, so it always represents a plane; varying λ tilts the plane around the fixed hinge ℓ.

PROPERTY **Family of planes through a line**

If $\Pi_1 : a_1x + b_1y + c_1z = d_1$ and $\Pi_2 : a_2x + b_2y + c_2z = d_2$ intersect in a line ℓ, then every plane through ℓ (except Π_2 itself) has an equation of the form

$$(a_1x + b_1y + c_1z - d_1) + \lambda(a_2x + b_2y + c_2z - d_2) = 0, \qquad \lambda \in \mathbb{R}.$$

(Proof not examinable; the justification above suffices.)

METHOD **Planes through the intersection of two planes**

If $\Pi_1 : a_1x + b_1y + c_1z = d_1$ and $\Pi_2 : a_2x + b_2y + c_2z = d_2$ intersect in a line ℓ, then every plane through ℓ (except Π_2 itself) has equation

$$(a_1x + b_1y + c_1z - d_1) + \lambda(a_2x + b_2y + c_2z - d_2) = 0, \qquad \lambda \in \mathbb{R}.$$

1. Write the family $\Pi_1 + \lambda\Pi_2 = 0$.
2. Use the extra condition (passing through a given point, perpendicular to a given plane, at a given distance from a point) to form an equation in λ.
3. Solve for λ and substitute back; simplify.
4. If the required plane turns out to be Π_2 itself, check it separately.

Reflex: any point of ℓ automatically satisfies the family equation for every λ — that is why the family consists exactly of the planes through ℓ.

WORKED EXAMPLE 28 **Plane through a line of intersection and an extra point**

Find the equation of the plane which passes through the line of intersection of the planes

$$\Pi_1 : x + y + z = 4 \qquad \text{and} \qquad \Pi_2 : 2x - y + z = 5,$$

and also passes through the point $Q(1, 2, 3)$.

SOLUTION

Step 1. The family. Every plane through the line of intersection has equation

$$(x + y + z - 4) + \lambda(2x - y + z - 5) = 0.$$

Step 2. Impose passage through $Q(1, 2, 3)$.

$$(1 + 2 + 3 - 4) + \lambda(2 - 2 + 3 - 5) = 0$$

$$2 + \lambda(-2) = 0 \implies \lambda = 1.$$

Step 3. Substitute back.

$$(x + y + z - 4) + (2x - y + z - 5) = 0$$

$$3x + 2z - 9 = 0, \qquad \text{i.e. } \Pi : 3x + 2z = 9.$$

Step 4. Verification.

- $Q(1, 2, 3)$: $3(1) + 2(3) = 9$ ✓.
- A point of the line of intersection, e.g. $(1, 0, 3)$ found earlier: $3(1) + 2(3) = 9$ ✓. Another point of the line, with $\lambda = 1$: $(3, 1, 0)$: $3(3) + 2(0) = 9$ ✓.

Hence the required plane is $3x + 2z = 9$.

WORKED EXAMPLE 29 **Plane of the family perpendicular to a given plane**

Find the equation of the plane through the line of intersection of $\Pi_1 : x + y + z = 4$ and $\Pi_2 : 2x - y + z = 5$ which is perpendicular to the plane $\Pi_3 : x - y + z = 0$.

SOLUTION

Step 1. The family is $(x + y + z - 4) + \lambda(2x - y + z - 5) = 0$, i.e.

$$(1 + 2\lambda)x + (1 - \lambda)y + (1 + \lambda)z - (4 + 5\lambda) = 0.$$

A normal vector of a plane of the family is $\vec{n}_\lambda = \begin{pmatrix} 1 + 2\lambda \\ 1 - \lambda \\ 1 + \lambda \end{pmatrix}$.

Step 2. Perpendicularity with Π_3 (normal $\vec{n}_3 = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$) requires $\vec{n}_\lambda \cdot \vec{n}_3 = 0$:

$$(1 + 2\lambda)(1) + (1 - \lambda)(-1) + (1 + \lambda)(1) = 0$$

$$1 + 2\lambda - 1 + \lambda + 1 + \lambda = 0 \implies 4\lambda + 1 = 0 \implies \lambda = -\tfrac{1}{4}.$$

Step 3. Substitute $\lambda = -\frac{1}{4}$:

$$\left(1 - \tfrac{1}{2}\right)x + \left(1 + \tfrac{1}{4}\right)y + \left(1 - \tfrac{1}{4}\right)z - \left(4 - \tfrac{5}{4}\right) = 0$$

$$\tfrac{1}{2}x + \tfrac{5}{4}y + \tfrac{3}{4}z - \tfrac{11}{4} = 0.$$

Multiplying by 4:

$$\Pi: \; 2x + 5y + 3z = 11.$$

Check: $\vec{n} \cdot \vec{n}_3 = 2 - 5 + 3 = 0$ ✓; the point $(1, 0, 3)$ of the line of intersection: $2 + 0 + 9 = 11$ ✓.

9.8 Method 8: Synthesis — a Complete GCE-Style Problem

METHOD **Tackling a multi-part vector geometry question**

1. Read the whole question first: later parts often reuse earlier results (normal vectors, intersection lines).
2. Keep exact values (fractions, surds) throughout; round only at the final answer.
3. After each computation, verify by substitution — examiners award marks for correct method even after a slip, but a check can save you.
4. Structure your answer with clear part labels (a), (b), (c)… as in the question.

WORKED EXAMPLE 30 **GCE-style synthesis**

The points $A(1, 1, 1)$, $B(2, 0, 3)$ and $C(0, 2, 2)$ define a plane Π, and the line L has equation $\vec{r} = \begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix} + t\begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$.

(a) Find the Cartesian equation of Π.

(b) Find the position vector of the point T where L meets Π.

(c) Find the angle between L and Π.

(d) Find the perpendicular distance from $D(4,0,1)$ to Π.

SOLUTION

(a) $\vec{AB} = \begin{pmatrix}1\\-1\\2\end{pmatrix}, \vec{AC} = \begin{pmatrix}-1\\1\\1\end{pmatrix}$.

$$\vec{n} = \vec{AB} \times \vec{AC} = \begin{pmatrix}(-1)(1)-(2)(1)\\(2)(-1)-(1)(1)\\(1)(1)-(-1)(-1)\end{pmatrix} = \begin{pmatrix}-3\\-3\\0\end{pmatrix} = -3\begin{pmatrix}1\\1\\0\end{pmatrix}.$$

Take $\vec{n} = \begin{pmatrix}1\\1\\0\end{pmatrix}$; then $d = 1+1+0 = 2$, so

$$\Pi: \; x + y = 2.$$

Check B: $2+0=2$ ✓; check C: $0+2=2$ ✓.

(b) On L: $x = 3+t$, $y = 1+t$, $z = -t$. Substitute into $x+y=2$:

$$(3+t)+(1+t) = 2 \implies 4+2t = 2 \implies t = -1.$$

Hence $T = (3-1,\ 1-1,\ 1) = (2,0,1)$, i.e. $\vec{OT} = \begin{pmatrix}2\\0\\1\end{pmatrix}$.

(c) With $\vec{d} = \begin{pmatrix}1\\1\\-1\end{pmatrix}$ and $\vec{n} = \begin{pmatrix}1\\1\\0\end{pmatrix}$:

$$\vec{d}\cdot\vec{n} = 1+1+0 = 2, \quad |\vec{d}| = \sqrt{3}, \quad |\vec{n}| = \sqrt{2}.$$

$$\sin\varphi = \frac{|2|}{\sqrt{3}\sqrt{2}} = \frac{2}{\sqrt{6}} = \frac{\sqrt{6}}{3} \approx 0.8165 \implies \varphi \approx 54.7^\circ.$$

(d) For $D(4,0,1)$:

$$\delta = \frac{|4+0-2|}{\sqrt{1^2+1^2+0^2}} = \frac{2}{\sqrt{2}} = \sqrt{2} \approx 1.41 \text{ units}.$$

KEY POINTS **Summary of key formulas**

- Plane through A with normal $\vec{n}$: $\vec{r}\cdot\vec{n} = \vec{a}\cdot\vec{n}$, i.e. $ax + by + cz = d$.
- Plane through A with directions $\vec{u}, \vec{v}$: $\vec{r} = \vec{a} + \lambda\vec{u} + \mu\vec{v}$; normal $\vec{n} = \vec{u}\times\vec{v}$.
- Distance from $P(x_1, y_1, z_1)$ to $ax + by + cz = d$: $\delta = \dfrac{|ax_1 + by_1 + cz_1 - d|}{\sqrt{a^2 + b^2 + c^2}}$.
- Angle between planes: $\cos\theta = \dfrac{|\vec{n}_1\cdot\vec{n}_2|}{|\vec{n}_1||\vec{n}_2|}$; angle line–plane: $\sin\varphi = \dfrac{|\vec{d}\cdot\vec{n}|}{|\vec{d}||\vec{n}|}$.
- Line of intersection of two planes: solve simultaneously; direction $\vec{n}_1\times\vec{n}_2$.
- Family through the line of intersection: $\Pi_1 + \lambda\Pi_2 = 0$.

WARNING **Final reminders for the examination**

- Never confuse the roles: for a plane, the normal gives the Cartesian coefficients; for a line, the direction vector gives the parametric coefficients.
- Angle line–plane: use $\sin\varphi$ with the normal — using $\cos$ gives the angle with the normal instead, a classic error.
- In the distance formula, do not forget the absolute value and the subtraction of d.
- When using the family $\Pi_1 + \lambda\Pi_2 = 0$, the plane Π_2 itself is not included; verify separately if needed.

EXERCISE 7 **Practice 1**

Find the Cartesian equation of the plane through $A(1, -2, 3)$ with normal vector $\vec{n} = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$.

SOLUTION **Practice 1**

The equation is $2x - y + 3z = d$ with $d = 2(1) - (-2) + 3(3) = 2 + 2 + 9 = 13$. Hence $\Pi:\ 2x - y + 3z = 13$.

EXERCISE 8 **Practice 2**

Find the Cartesian equation of the plane through $A(1, 1, 1)$, $B(2, 3, 0)$ and $C(-1, 0, 2)$.

SOLUTION **Practice 2**

$\vec{AB} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$, $\vec{AC} = \begin{pmatrix} -2 \\ -1 \\ 1 \end{pmatrix}$ (not parallel, so the points are not collinear).

$$\vec{n} = \vec{AB} \times \vec{AC} = \begin{pmatrix} (2)(1) - (-1)(-1) \\ (-1)(-2) - (1)(1) \\ (1)(-1) - (2)(-2) \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix}.$$

Then $d = 1 + 1 + 3 = 5$, so $\Pi: x + y + 3z = 5$. Check B: $2 + 3 + 0 = 5$ ✓; check C: $-1 + 0 + 6 = 5$ ✓.

EXERCISE 9 **Practice 3**

Find the perpendicular distance from $P(2, 0, -1)$ to the plane $\Pi: x - 2y + 2z = 8$.

SOLUTION **Practice 3**

$$\delta = \frac{|(2) - 2(0) + 2(-1) - 8|}{\sqrt{1 + 4 + 4}} = \frac{|2 - 2 - 8|}{3} = \frac{8}{3} \text{ units.}$$

EXERCISE 10 **Practice 4**

Find the acute angle between the planes $x + y + z = 1$ and $x - y = 0$.

SOLUTION **Practice 4**

$\vec{n}_1 \cdot \vec{n}_2 = 1 - 1 + 0 = 0$: the normals are perpendicular, so the planes are perpendicular; the angle is 90°.

EXERCISE 11 **Practice 5**

Find the line of intersection of the planes $x + y - z = 2$ and $2x - y + z = 1$, giving your answer in vector form.

SOLUTION **Practice 5**

Add the equations: $3x = 3$, so $x = 1$. Then from the first: $z = x + y - 2 = y - 1$. Let $y = \lambda$: $z = \lambda - 1$. Hence

$$\vec{r} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}.$$

Check direction: $\vec{n}_1 \times \vec{n}_2 = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} \times \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1-1 \\ -2-1 \\ -1-2 \end{pmatrix} = \begin{pmatrix} 0 \\ -3 \\ -3 \end{pmatrix} = -3 \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$ ✓.

EXERCISE 12 **Practice 6**

Find the equation of the plane through the line of intersection of $x+2y-z = 3$ and $2x-y+z = 4$ which passes through the origin.

SOLUTION **Practice 6**

Family: $(x + 2y - z - 3) + \lambda(2x - y + z - 4) = 0$. At $(0, 0, 0)$: $-3 - 4\lambda = 0$, so $\lambda = -\frac{3}{4}$. Substituting:

$$(x + 2y - z - 3) - \tfrac{3}{4}(2x - y + z - 4) = 0 \implies -\tfrac{1}{2}x + \tfrac{11}{4}y - \tfrac{7}{4}z = 0.$$

Multiplying by -4: $2x - 11y + 7z = 0$. Check: the origin satisfies it ✓, and it contains the line of intersection by construction.

DID YOU KNOW? **Why planes matter beyond the classroom**

Vector equations of planes are the working tool of 3D computer graphics: every flat face of an object on a screen is stored as a plane, and the rendering engine computes line–plane intersections and angles between planes thousands of times per second to decide what is visible and how light reflects. The same mathematics is used by civil engineers — for instance when checking the inclination of a roof or the angle between two road surfaces —

and by geologists studying the tilt of rock layers beneath the soil of the Cameroon Volcanic Line.

10 EXERCISES

10.1 I check my knowledge

EXERCISE 13 **Multiple choice**

Choose the correct answer, justifying your choice briefly.

1. A Cartesian equation of the plane through $A(1, -2, 3)$ with normal vector $\vec{n} = \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}$ is:

   (a) $2x + y - z = 3$

   (b) $2x + y - z = -3$

   (c) $x - 2y + 3z = 0$

   (d) $2x + y - z = 7$

2. The planes $x - 2y + z = 4$ and $2x - 4y + 2z = 9$ are:

   (a) identical

   (b) intersecting along a line

   (c) parallel and distinct

   (d) perpendicular

3. The perpendicular distance from the origin to the plane $3x + 4y + 12z = 26$ is:

   (a) 1

   (b) 2

   (c) 13

   (d) 26

4. The line $\vec{r} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ and the plane $x - y + z = 5$:

   (a) meet at exactly one point

   (b) are parallel, the line not lying in the plane

   (c) are such that the line lies in the plane

   (d) are perpendicular

EXERCISE 14 **True or false**

State whether each assertion is true or false, justifying your answer.

1. If three points are collinear, then there is exactly one plane passing through them.
2. The angle between two planes is equal to the angle between their normal vectors (taken acute).
3. If $\vec{n}$ is a normal vector of a plane Π, then $3\vec{n}$ is also a normal vector of Π.
4. If a line is perpendicular to a plane, then its direction vector is parallel to every vector lying in the plane.

EXERCISE 15 **Definitions and vocabulary**

1. Define a **normal vector** of a plane and explain why a plane has infinitely many normal vectors.
2. State the vector equation of a plane through a point A, parallel to two non-parallel vectors $\vec{u}$ and $\vec{v}$.
3. State the formula giving the perpendicular distance from the point $P(x_0, y_0, z_0)$ to the plane $ax + by + cz = d$, and say what quantity appears in the denominator.

EXERCISE 16 **Direct calculations**

1. Write down a normal vector of each plane: (i) $2x - 3y + z = 7$; (ii) $x + 5z = 0$.
2. Find the acute angle between the planes $x + y + z = 1$ and $x - y = 0$, giving your answer to the nearest degree.
3. Find the perpendicular distance from the point $B(2, -1, 4)$ to the plane $2x - y + 2z = 6$.

10.2 I practise

EXERCISE 17 **Cartesian equation and position of points**

1. Find the Cartesian equation of the plane through $A(1, 2, -1)$ with normal vector $2\vec{i} - \vec{j} + 3\vec{k}$.
2. Determine whether the points $(3, 1, 0)$ and $(0, 5, 1)$ lie on the same side of this plane.

EXERCISE 18 **Plane through three points**

1. Verify that the points $P(1, 0, 2)$, $Q(2, 1, 1)$ and $R(-1, 2, 0)$ are not collinear.
2. Find the Cartesian equation of the plane through P, Q and R.

EXERCISE 19 **From parametric to Cartesian form**

A plane has parametric (vector) equation

$$\vec{r} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix}, \qquad \lambda, \mu \in \mathbb{R}.$$

1. Find a normal vector of the plane and deduce its Cartesian equation.
2. Find the perpendicular distance from the origin to the plane.

EXERCISE 20 **Angle between two planes**

Find the acute angle between the planes $2x + y - 2z = 5$ and $3x - 6y - 2z = 7$, giving your answer to the nearest degree.

10.3 I prepare for the GCE

EXERCISE 21 **Line and plane (8 marks)**

The line l has equation $\vec{r} = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$ and the plane Π has equation $x - y + 2z = 4$.

1. Find the coordinates of the point where l meets Π. (3 marks)
2. Find the acute angle between the line l and the plane Π, to the nearest degree. (3 marks)
3. Determine whether the point $(4, 3, 1)$ lies on l, and whether it lies in Π. (2 marks)

EXERCISE 22 **Line parallel to a plane; line of intersection (9 marks)**

1. Show that the line $\vec{r} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}$ is parallel to the plane $2x - 2y + z = 7$, and find the distance between the line and the plane. (4 marks)

2. Find the vector equation of the line of intersection of the planes $x+y+z = 6$ and $2x-y+z = 3$, and check that the point $(1, 2, 3)$ lies on this line. (5 marks)

EXERCISE 23 **Families of planes; extension (10 marks)**

1. Find the equation of the plane which passes through the line of intersection of the planes $x + 2y - z = 4$ and $2x - y + z = 1$ and also passes through the point $(1, 1, 2)$. (4 marks)

2. Find the equation of the plane through the line of intersection of $x + y - z = 2$ and $2x - 3y + z = 5$ which is perpendicular to the plane $x - y + z = 0$. (4 marks)

3. (For strong candidates.) The planes $2x - y + 2z = 8$ and $4x - 2y + 4z = 3$ are given. Show that they are parallel, find the distance between them, and deduce the equation of the plane midway between them. (2 marks)

11 SOLUTIONS TO THE EXERCISES

11.1 I check my knowledge

SOLUTION **Exercise 1 — Multiple choice**

1. Answer (b): $2x + y - z = -3$.

   A plane with normal vector $\vec{n} = \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}$ has equation $2x + y - z = d$. Since $A(1, -2, 3)$ lies on it, $d = 2(1) + (-2) - (3) = -3$. Hence $2x + y - z = -3$.

2. Answer (c): parallel and distinct.

   The normal vectors are $\vec{n}_1 = (1, -2, 1)$ and $\vec{n}_2 = (2, -4, 2) = 2\vec{n}_1$: the normals are parallel, so the planes are parallel. But the right-hand sides do not scale the same way ($2 \times 4 = 8 \neq 9$), so the planes are distinct.

3. Answer (b): 2.

   Distance from the origin: $\delta = \dfrac{|-26|}{\sqrt{3^2 + 4^2 + 12^2}} = \dfrac{26}{\sqrt{169}} = \dfrac{26}{13} = 2.$

4. Answer (b): parallel, the line not lying in the plane.

   Direction $\vec{d} = (1, 1, 1)$, normal $\vec{n} = (1, -1, 1)$: $\vec{d} \cdot \vec{n} = 1-1+1 = 1 \neq 0$? Let us recompute: $1-1+1 = 1$. Hmm — check whether the line meets the plane: substitute $x = 1 + \lambda$, $y = \lambda$, $z = 2 + \lambda$ into $x - y + z = 5$: $(1 + \lambda) - \lambda + (2 + \lambda) = 3 + \lambda = 5$, so $\lambda = 2$. That gives a unique intersection point $(3, 2, 4)$, which would be answer (a).

   Let us recheck $\vec{d} \cdot \vec{n} = 1(1) + 1(-1) + 1(1) = 1 \neq 0$, so the line is not parallel to the plane and meets it in exactly one point. Correct answer: (a) — the line meets the plane at the single point $(3, 2, 4)$ (obtained for $\lambda = 2$).

SOLUTION **Exercise 2 — True or false**

1. False. If the three points are collinear, infinitely many planes contain the line through them (any plane containing that line). Uniqueness requires non-collinear points.

2. True. By definition, the (acute) angle between two planes is the acute angle between their normal vectors: $\cos\theta = \dfrac{|\vec{n}_1 \cdot \vec{n}_2|}{\|\vec{n}_1\| \, \|\vec{n}_2\|}$.

3. True. A normal vector is any non-zero vector perpendicular to the plane; any non-zero

scalar multiple of $\vec{n}$ is still perpendicular to every direction of the plane.

4. False. If a line is perpendicular to a plane, its direction vector is parallel to the normal of the plane, hence perpendicular to every vector lying in the plane — not parallel.

SOLUTION **Exercise 3 — Definitions and vocabulary**

1. A **normal vector** of a plane Π is any non-zero vector perpendicular to every vector lying in Π (equivalently, perpendicular to two non-parallel directions of Π). If $\vec{n}$ is normal to Π, so is $k\vec{n}$ for every $k \neq 0$; hence a plane has infinitely many normal vectors, all collinear.
2. Vector equation: $\vec{r} = \vec{OA} + \lambda\vec{u} + \mu\vec{v}, \lambda, \mu \in \mathbb{R}$, where $\vec{u}$ and $\vec{v}$ are non-parallel.
3. $\delta = \dfrac{|ax_0 + by_0 + cz_0 - d|}{\sqrt{a^2 + b^2 + c^2}}$. The denominator is the norm $\|\vec{n}\|$ of a normal vector $\vec{n} = (a, b, c)$ of the plane.

SOLUTION **Exercise 4 — Direct calculations**

1. (i) $\vec{n} = (2, -3, 1)$; (ii) $\vec{n} = (1, 0, 5)$ (note the missing y-term gives coefficient 0).
2. Normals: $\vec{n}_1 = (1, 1, 1)$, $\vec{n}_2 = (1, -1, 0)$.

$$\cos\theta = \frac{|\vec{n}_1 \cdot \vec{n}_2|}{\|\vec{n}_1\|\,\|\vec{n}_2\|} = \frac{|1 - 1 + 0|}{\sqrt{3}\sqrt{2}} = 0 \quad \Longrightarrow \quad \theta = 90^\circ.$$

   The planes are perpendicular.

3. $\delta = \dfrac{|2(2) - (-1) + 2(4) - 6|}{\sqrt{4 + 1 + 4}} = \dfrac{|4 + 1 + 8 - 6|}{3} = \dfrac{7}{3} \approx 2.33$ units.

11.2 I practise

SOLUTION **Exercise 5 — Cartesian equation and position of points**

1. The plane has equation $2x - y + 3z = d$. Substituting $A(1, 2, -1)$: $d = 2 - 2 - 3 = -3$. Hence

$$\boxed{2x - y + 3z = -3}.$$

2. Define $f(x,y,z) = 2x - y + 3z + 3$. For $(3,1,0)$: $f = 6 - 1 + 0 + 3 = 8 > 0$. For $(0,5,1)$: $f = 0 - 5 + 3 + 3 = 1 > 0$. Both values have the same sign, so the two points lie on the same side of the plane.

SOLUTION **Exercise 6 — Plane through three points**

1. $\vec{PQ} = (1,1,-1)$ and $\vec{PR} = (-2,2,-2)$. These vectors are not proportional ($\frac{1}{-2} \neq \frac{1}{2}$), so P, Q, R are not collinear.

2. Normal vector:

$$\vec{n} = \vec{PQ} \times \vec{PR} = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} \times \begin{pmatrix} -2 \\ 2 \\ -2 \end{pmatrix} = \begin{pmatrix} (1)(-2) - (-1)(2) \\ (-1)(-2) - (1)(-2) \\ (1)(2) - (1)(-2) \end{pmatrix} = \begin{pmatrix} 0 \\ 4 \\ 4 \end{pmatrix}.$$

Take the simpler normal $(0,1,1)$. Equation: $y + z = d$; using $P(1,0,2)$: $d = 2$. Hence

$$\boxed{y + z = 2}.$$

Check: Q: $1 + 1 = 2$ ✓; R: $2 + 0 = 2$ ✓.

SOLUTION **Exercise 7 — From parametric to Cartesian form**

1. Direction vectors $\vec{u} = (1,1,0)$, $\vec{v} = (0,2,1)$. A normal vector is

$$\vec{n} = \vec{u} \times \vec{v} = \begin{pmatrix} (1)(1) - (0)(2) \\ (0)(0) - (1)(1) \\ (1)(2) - (1)(0) \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}.$$

Equation: $x - y + 2z = d$; the point $(1,0,2)$ lies in the plane, so $d = 1 + 4 = 5$:

$$\boxed{x - y + 2z = 5}.$$

2. Distance from the origin:

$$\delta = \frac{|0 - 0 + 0 - 5|}{\sqrt{1 + 1 + 4}} = \frac{5}{\sqrt{6}} = \frac{5\sqrt{6}}{6} \approx 2.04 \text{ units}.$$

SOLUTION **Exercise 8 — Angle between two planes**

Normals: $\vec{n}_1 = (2, 1, -2)$, $\vec{n}_2 = (3, -6, -2)$.

$$\vec{n}_1 \cdot \vec{n}_2 = 6 - 6 + 4 = 4, \qquad \|\vec{n}_1\| = \sqrt{4 + 1 + 4} = 3, \qquad \|\vec{n}_2\| = \sqrt{9 + 36 + 4} = 7.$$

$$\cos\theta = \frac{|4|}{3 \times 7} = \frac{4}{21} \approx 0.1905 \quad \Longrightarrow \quad \theta \approx 79^\circ \text{ (nearest degree).}$$

11.3 I prepare for the GCE

SOLUTION **Exercise 9 — Line and plane (8 marks)**

1. Parametrize: $x = 2 + \lambda$, $y = -1 + 2\lambda$, $z = 3 - \lambda$. Substitute into $x - y + 2z = 4$:

   $$(2 + \lambda) - (-1 + 2\lambda) + 2(3 - \lambda) = 4 \implies 2 + \lambda + 1 - 2\lambda + 6 - 2\lambda = 4 \implies 9 - 3\lambda = 4,$$

   so $\lambda = \dfrac{5}{3}$. The intersection point is

   $$\left(2 + \tfrac{5}{3},\ -1 + \tfrac{10}{3},\ 3 - \tfrac{5}{3}\right) = \boxed{\left(\tfrac{11}{3},\ \tfrac{7}{3},\ \tfrac{4}{3}\right)}.$$

   Mark scheme: M1 substitution; A1 value of λ; A1 point.

2. Direction $\vec{d} = (1, 2, -1)$, normal $\vec{n} = (1, -1, 2)$.

   $$\vec{d} \cdot \vec{n} = 1 - 2 - 2 = -3, \quad \|\vec{d}\| = \sqrt{6}, \quad \|\vec{n}\| = \sqrt{6}.$$

   If φ is the angle between $\vec{d}$ and $\vec{n}$ and θ the angle between line and plane, $\sin\theta = |\cos\varphi|$:

   $$\sin\theta = \frac{|\vec{d} \cdot \vec{n}|}{\|\vec{d}\|\,\|\vec{n}\|} = \frac{3}{6} = \frac{1}{2} \quad \Longrightarrow \quad \boxed{\theta = 30^\circ}.$$

   Mark scheme: M1 correct scalar product; M1 use of $\sin\theta$ formula; A1 answer. Examiner expects the acute angle between line and plane, i.e. the complement of the angle with the normal.

3. On l: we need $2 + \lambda = 4$, $-1 + 2\lambda = 3$, $3 - \lambda = 1$, i.e. $\lambda = 2$ from all three equations — consistent. So $(4, 3, 1)$ lies on l.

   In Π: $4 - 3 + 2(1) = 3 \neq 4$, so $(4, 3, 1)$ does not lie in Π.

   Mark scheme: B1 for each correct conclusion with justification.

SOLUTION **Exercise 10 — Line parallel to a plane; line of intersection (9 marks)**

1. Direction $\vec{d} = (2, 1, -2)$, normal $\vec{n} = (2, -2, 1)$:

$$\vec{d} \cdot \vec{n} = 4 - 2 - 2 = 0,$$

so the line is parallel to the plane (or lies in it). Test the point $(1, 1, 0)$ of the line: $2 - 2 + 0 = 0 \neq 7$, so the line does not lie in the plane: it is strictly parallel.

The distance between line and plane equals the distance from any point of the line, e.g. $(1, 1, 0)$, to the plane:

$$\delta = \frac{|2(1) - 2(1) + 0 - 7|}{\sqrt{4 + 4 + 1}} = \boxed{\frac{7}{3} \approx 2.33 \text{ units}}.$$

Mark scheme: M1 scalar product; A1 conclusion parallel; B1 check the line is not in the plane; A1 distance.

2. Solve the system

$$\begin{cases} x + y + z = 6 \\ 2x - y + z = 3 \end{cases}$$

Set $z = \lambda$. Subtracting the first equation from the second: $x - 2y = -3$, so $x = 2y - 3$. Substituting into the first: $(2y - 3) + y + \lambda = 6$, so $3y = 9 - \lambda$, $y = 3 - \frac{\lambda}{3}$, and $x = 2\left(3 - \frac{\lambda}{3}\right) - 3 = 3 - \frac{2\lambda}{3}$.

To avoid fractions, put $\lambda = 3t$: then

$$x = 3 - 2t, \qquad y = 3 - t, \qquad z = 3t,$$

i.e.

$$\boxed{\vec{r} = \begin{pmatrix} 3 \\ 3 \\ 0 \end{pmatrix} + t \begin{pmatrix} -2 \\ -1 \\ 3 \end{pmatrix}, \quad t \in \mathbb{R}.}$$

Check direction: $(-2, -1, 3)$ is perpendicular to both normals: $(-2)(1) + (-1)(1) + (3)(1) = 0$ and $(-2)(2) + (-1)(-1) + (3)(1) = -4 + 1 + 3 = 0$ ✓.

Check $(1, 2, 3)$: $3 - 2t = 1 \Rightarrow t = 1$; then $y = 3 - 1 = 2$ ✓, $z = 3$ ✓. Also $1 + 2 + 3 = 6$ and $2 - 2 + 3 = 3$ ✓. So $(1, 2, 3)$ lies on the line.

Mark scheme: M1 elimination; A1 parametric solution; A1 vector equation; M1 verification method; A1 conclusion.

SOLUTION **Exercise 11 — Families of planes; extension (10 marks)**

1. Any plane through the line of intersection of $x + 2y - z = 4$ and $2x - y + z = 1$ has equation

$$(x + 2y - z - 4) + \lambda(2x - y + z - 1) = 0.$$

Impose passage through $(1, 1, 2)$:

$$(1 + 2 - 2 - 4) + \lambda(2 - 1 + 2 - 1) = 0 \implies -3 + 2\lambda = 0 \implies \lambda = \tfrac{3}{2}.$$

Hence

$$(x + 2y - z - 4) + \tfrac{3}{2}(2x - y + z - 1) = 0 \implies 2(x + 2y - z - 4) + 3(2x - y + z - 1) = 0,$$

$$\boxed{8x + y + z - 11 = 0}.$$

Mark scheme: M1 family of planes; M1 substitution of the point; A1 λ; A1 equation.

2. Family: $(x + y - z - 2) + \lambda(2x - 3y + z - 5) = 0$, i.e.

$$(1 + 2\lambda)x + (1 - 3\lambda)y + (-1 + \lambda)z - (2 + 5\lambda) = 0.$$

Its normal is $\vec{n} = (1 + 2\lambda,\ 1 - 3\lambda,\ -1 + \lambda)$. Perpendicularity to $x - y + z = 0$ (normal $(1, -1, 1)$) requires $\vec{n} \cdot (1, -1, 1) = 0$:

$$(1 + 2\lambda) - (1 - 3\lambda) + (-1 + \lambda) = 0 \implies 6\lambda - 1 = 0 \implies \lambda = \tfrac{1}{6}.$$

Substituting:

$$\left(1 + \tfrac{1}{3}\right)x + \left(1 - \tfrac{1}{2}\right)y + \left(-1 + \tfrac{1}{6}\right)z - \left(2 + \tfrac{5}{6}\right) = 0 \implies \tfrac{4}{3}x + \tfrac{1}{2}y - \tfrac{5}{6}z - \tfrac{17}{6} = 0.$$

Multiplying by 6: $\boxed{8x + 3y - 5z - 17 = 0}$.

Mark scheme: M1 family; M1 perpendicularity condition on normals; A1 λ; A1 equation.

3. Normals: $\vec{n}_1 = (2, -1, 2)$ and $\vec{n}_2 = (4, -2, 4) = 2\vec{n}_1$: the planes are parallel. They are distinct since $\dfrac{8}{2} = 4 \neq \dfrac{3}{2}$ (the equations are not proportional).

Distance: take a point of the first plane, e.g. $(4, 0, 0)$ (check: $8 = 8$ ✓), and compute its distance to $4x - 2y + 4z = 3$:

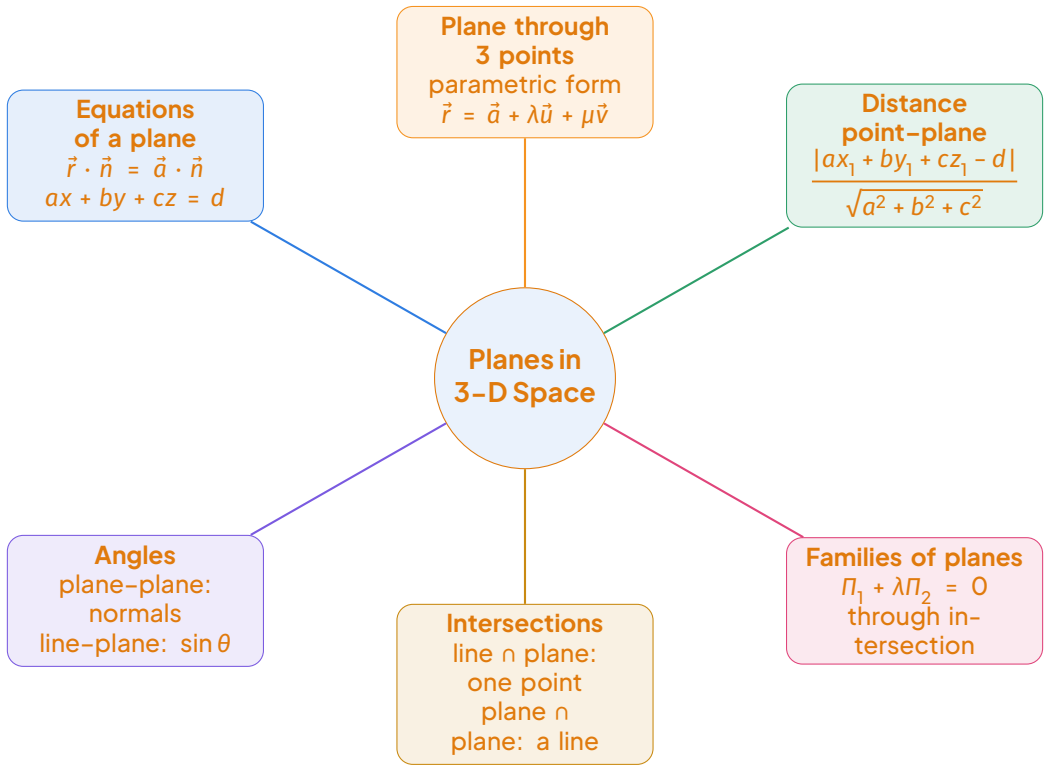
$$\delta = \frac{|4(4) - 0 + 0 - 3|}{\sqrt{16 + 4 + 16}} = \frac{13}{6} \approx 2.17 \text{ units}.$$

Midway plane: it is parallel to both, so has equation $2x - y + 2z = k$ with k the mean of the right-hand sides written with the same coefficients. Writing the second plane as $2x - y + 2z = \dfrac{3}{2}$:

$$k = \frac{8 + \frac{3}{2}}{2} = \frac{19}{4}, \qquad \text{so} \qquad \boxed{2x - y + 2z = \tfrac{19}{4}} \text{ or equivalently } 8x - 4y + 8z = 19.$$

Mark scheme: B1 parallel + distance; B1 midway plane. Examiner expects candidates to normalise the equations before averaging.

12 SUMMARY SHEET



Mind map of the chapter Vectors: planes in three dimensions.

FICHE BILAN **L'essentiel à connaître pour l'examen**

Key definitions.

- A **plane** is determined by a point A and a **normal vector** $\vec{n}$, or by three non-collinear points, or by a point and two non-parallel direction vectors.
- **Vector equation** of a plane: $(\vec{r} - \vec{a}) \cdot \vec{n} = 0$, i.e. $\vec{r} \cdot \vec{n} = \vec{a} \cdot \vec{n}$.
- **Cartesian equation**: $ax + by + cz = d$, where $\vec{n} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$ is a normal vector and $d = \vec{a} \cdot \vec{n}$.
- **Parametric form**: $\vec{r} = \vec{a} + \lambda\vec{u} + \mu\vec{v}$, $\lambda, \mu \in \mathbb{R}$, with $\vec{u}, \vec{v}$ non-parallel vectors lying in the plane.

Formulas and results to know.

- Plane through $A(x_1, y_1, z_1)$ with normal (a, b, c): $a(x - x_1) + b(y - y_1) + c(z - z_1) = 0$.
- Plane through three points A, B, C: direction vectors $\vec{AB}$, $\vec{AC}$; normal $\vec{n} = \vec{AB} \times \vec{AC}$ (the

three points must not be collinear).

- **Perpendicular distance** from $P(x_1, y_1, z_1)$ to $ax + by + cz = d$:

$$\delta = \frac{|ax_1 + by_1 + cz_1 - d|}{\sqrt{a^2 + b^2 + c^2}}.$$

- **Angle between two planes** = angle between their normals:

$$\cos\theta = \frac{|\vec{n_1} \cdot \vec{n_2}|}{|\vec{n_1}|\,|\vec{n_2}|}.$$

- **Angle between a line** (direction $\vec{u}$) and a plane (normal $\vec{n}$):

$$\sin\theta = \frac{|\vec{u} \cdot \vec{n}|}{|\vec{u}|\,|\vec{n}|}.$$

- **Intersection line of two planes**: direction $\vec{u} = \vec{n_1} \times \vec{n_2}$; find one common point by fixing one coordinate.
- **Family of planes** through the line of intersection of $\Pi_1 = 0$ and $\Pi_2 = 0$: $\Pi_1 + \lambda\Pi_2 = 0$ (plus Π_2 itself).

Methods.

- Line–plane intersection: substitute the parametric equations of the line into the plane equation; solve for the parameter; substitute back. No solution ⇒ line parallel to plane; infinitely many ⇒ line lies in the plane.
- To find a plane through the intersection of two planes and satisfying one extra condition (a point, perpendicularity), write $\Pi_1 + \lambda\Pi_2 = 0$ and determine λ.
- To convert parametric to Cartesian form, compute $\vec{n} = \vec{u} \times \vec{v}$ then $d = \vec{a} \cdot \vec{n}$.

Common mistakes.

- Forgetting the absolute value in distance and angle formulas (distances and acute angles are non-negative).
- Using $\cos\theta$ instead of $\sin\theta$ for the line–plane angle: the line–plane angle is the complement of the line–normal angle.
- Trying to build a plane from three collinear points: infinitely many planes pass through them, so the plane is not unique.
- Confusing the normal vector with a direction vector of the plane: the normal is perpendicular to the plane.
- Omitting the case $\Pi_2 = 0$ when using the family $\Pi_1 + \lambda\Pi_2 = 0$.

GCE tips.

- Always state $\vec{n}$ explicitly; most marks in vector questions come from identifying normals correctly.
- Show the substitution step fully when intersecting a line and a plane; examiners award method marks.
- Check your final plane equation by substituting a known point of the plane.
- Keep exact values (fractions, surds) until the final answer, then round sensibly.

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